User Manual

for S32K3 RM Driver

Document Number: UM34RMASR4.4 Rev0000R2.0.0 Rev. 1.0

1 Revision History **2**

2 Introduction **3**

2.1 Supported Derivatives . . . 3
2.2 Overview . . . 4
2.3 About This Manual . . . 4
2.4 Acronyms and Definitions . . . 5
2.5 Reference List . . . 5

3 Driver **6**

3.1 Requirements . . . 6
3.2 Driver Design Summary . . . 6
3.3 Hardware Resources . . . 6
3.4 Deviations from Requirements . . . 7
3.5 Driver Limitations . . . 7
3.6 Driver usage and configuration tips . . . 7
3.6.1 HLD usage . . . 7
3.6.2 LLD usage . . . 8
3.6.3 Interface configuration . . . 9
3.6.4 Multicore initialization . . . 10
3.6.5 XRDC configuration . . . 10
3.6.6 Sema42 logic channel configuration . . . 15
3.6.7 MPU M7 configuration . . . 16
3.6.8 Errata for MPU M7 : [Core] Speculative accesses might be performed to memory unmapped in MPU. . . . 17
3.7 Runtime errors . . . 18
3.8 Symbolic Names Disclaimer . . . 19

4 Tresos Configuration Plug-in **20**

4.1 Module Rm . . . 24
4.2 Container RmGeneral . . . 24
4.3 Parameter Rm_VersionInfoApi . . . 24
4.4 Parameter RmDevErrorDetect . . . 25
4.5 Parameter RmEnableUserModeSupport . . . 25
4.6 Parameter RmEnableMultiCore . . . 26
4.7 Parameter RmEnableXRDCSupport . . . 26
4.8 Parameter RmEnableSema42Support . . . 27
4.9 Parameter RmMpuM7Configurable . . . 27
4.10 Parameter RmPflashConfigurable . . . 28
4.11 Parameter RmCrossbarConfigurable . . . 28
4.12 Parameter RmXbicConfigurable . . . 29

4.13 Parameter RmVirtWrapperConfigurable . . . 29
4.14 Container RmConfigSet . . . 30
4.15 Container Xrdc_Configuration . . . 31
4.16 Parameter XrdcDevErrorDetect . . . 31
4.17 Parameter XrdcRegistersLock . . . 31
4.18 Parameter XrdcPIDRegisterLock . . . 32
4.19 Container Rm_XRDC_Domain_Assignment . . . 32
4.20 Parameter DomainID . . . 33
4.21 Container Domain_Master_Assignment . . . 33
4.22 Parameter XrdcMasterInstance . . . 34
4.23 Parameter XrdcPIDEnable . . . 34
4.24 Parameter XrdcMasterPID . . . 35
4.25 Parameter XrdcMasterPIDMask . . . 35
4.26 Parameter XrdcDIDBDisable . . . 36
4.27 Parameter XrdcMasterMode . . . 36
4.28 Parameter XrdcMasterPriviledgeMode . . . 37
4.29 Container Domain_Peripheral_Assignment . . . 37
4.30 Parameter Secure_Access_Policy . . . 37
4.31 Parameter Non_Secure_Access_Policy . . . 40
4.32 Reference Domain_Peripheral_Assignment . . . 43
4.33 Container Domain_Memory_Assignment . . . 43
4.34 Parameter Secure_Access_Policy . . . 43
4.35 Parameter Non_Secure_Access_Policy . . . 46
4.36 Reference Domain_Memory_Assignment . . . 49
4.37 Container Rm_XRDC_Memory_Config . . . 49
4.38 Parameter XrdcMrcInstance . . . 49
4.39 Parameter XrdcStartAddress . . . 50
4.40 Parameter XrdcEndAddress . . . 50
4.41 Parameter XrdcSema4Enable . . . 51
4.42 Reference XrdcSema42LogicChannel . . . 51
4.43 Container Rm_XRDC_Peripheral_Config . . . 52
4.44 Parameter XrdcPeripheralSlot . . . 52
4.45 Parameter XrdcSema4Enable . . . 54
4.46 Reference XrdcSema42LogicChannel . . . 54
4.47 Container MPU_M7_Configuration . . . 54
4.48 Parameter MpuDevErrorDetect . . . 56
4.49 Parameter ClearCacheEnable . . . 56
4.50 Container MPU_M7_ModuleConfig . . . 57
4.51 Parameter DefaultMapEnable . . . 57
4.52 Parameter RunInHFNMIEnable . . . 58

4.53 Parameter MemManageInterruptEnable . . . 58
4.54 Parameter PhysicalCoreID . . . 58
4.55 Container RegionConfig . . . 59
4.56 Parameter RegionNumber . . . 59
4.57 Parameter StartAddress . . . 60
4.58 Parameter EndAddress . . . 60
4.59 Parameter RegionSize . . . 61
4.60 Parameter MemoryType . . . 61
4.61 Parameter AccessRights . . . 62
4.62 Parameter OuterCachePolicy . . . 63
4.63 Parameter InnerCachePolicy . . . 63
4.64 Parameter SubregionMask . . . 64
4.65 Parameter Shareable . . . 64
4.66 Container Sema42_ModuleConfig . . . 64
4.67 Parameter Sema42DevErrorDetect . . . 65
4.68 Container Sema42LogicChannelConfiguration . . . 65
4.69 Parameter Sema42LogicChannel_LogicName . . . 66
4.70 Parameter Sema42HardwareInstance . . . 66
4.71 Parameter Sema42HardwareChannel . . . 67
4.72 Container Pflash_Configuration . . . 67
4.73 Parameter PflashDevErrorDetect . . . 68
4.74 Container PflashMasterProtection . . . 68
4.75 Parameter PflashControler . . . 68
4.76 Parameter PflashMAP . . . 69
4.77 Parameter PflashMasterAccess . . . 69
4.78 Container Crossbar_Configuration . . . 70
4.79 Parameter CrossbarDevErrorDetect . . . 70
4.80 Container RmCrossbarInstance . . . 71
4.81 Parameter RmCrossbarHwInstance . . . 71
4.82 Container RmCrossbarHwSlavePort . . . 72
4.83 Parameter RmSlavePortNumber . . . 72
4.84 Parameter RmCrossbarPrioMaster0 . . . 73
4.85 Parameter RmCrossbarPrioMaster1 . . . 73
4.86 Parameter RmCrossbarPrioMaster2 . . . 74
4.87 Parameter RmCrossbarPrioMaster3 . . . 74
4.88 Parameter RmCrossbarPrioMaster4 . . . 75
4.89 Parameter RmCrossbarPrioMaster5 . . . 75
4.90 Parameter RmCrossbarPrioMaster6 . . . 76
4.91 Parameter RmCrossbarPrioMaster7 . . . 76
4.92 Parameter RmCrossbarEnableLock . . . 77

4.93 Parameter RmCrossbarHaltLowPrio . . . 77
4.94 Parameter RmCrossbarEnablePrioElevM0 . . . 78
4.95 Parameter RmCrossbarEnablePrioElevM1 . . . 78
4.96 Parameter RmCrossbarEnablePrioElevM2 . . . 79
4.97 Parameter RmCrossbarEnablePrioElevM3 . . . 79
4.98 Parameter RmCrossbarEnablePrioElevM4 . . . 80
4.99 Parameter RmCrossbarEnablePrioElevM5 . . . 80
4.100 Parameter RmCrossbarEnablePrioElevM6 . . . 81
4.101 Parameter RmCrossbarEnablePrioElevM7 . . . 81
4.102 Parameter RmCrossbarEnableFixedPrio . . . 82
4.103 Parameter RmCrossbarParkingControl . . . 82
4.104 Parameter RmCrossbarParkField . . . 83
4.105 Container Xbic_Configuration . . . 83
4.106 Parameter XbicDevErrorDetect . . . 84
4.107 Container XbicCheckingControl . . . 84
4.108 Parameter XbicInstance . . . 85
4.109 Parameter SE0 . . . 85
4.110 Parameter SE1 . . . 86
4.111 Parameter SE2 . . . 86
4.112 Parameter SE3 . . . 87
4.113 Parameter SE4 . . . 87
4.114 Parameter SE5 . . . 87
4.115 Parameter SE6 . . . 88
4.116 Parameter SE7 . . . 88
4.117 Parameter ME0 . . . 89
4.118 Parameter ME1 . . . 89
4.119 Parameter ME2 . . . 90
4.120 Parameter ME3 . . . 90
4.121 Parameter ME4 . . . 90
4.122 Parameter ME5 . . . 91
4.123 Parameter ME6 . . . 91
4.124 Parameter ME7 . . . 92
4.125 Container Virt_Wrapper_Configuration . . . 92
4.126 Parameter VirtWrapperDevErrorDetect . . . 93
4.127 Container Mscr_Config_List . . . 93
4.128 Parameter MscrName . . . 93
4.129 Parameter MscrNumber . . . 94
4.130 Parameter MscrSiul2Instance . . . 95
4.131 Parameter MscrPinMux . . . 95
4.132 Parameter MscrMirror . . . 96

4.133 Container Imcr_Config_List . . . 96
4.134 Parameter ImcrName . . . 97
4.135 Parameter ImcrNumber . . . 99
4.136 Parameter ImcrSiul2Instance . . . 99
4.137 Parameter ImcrInput . . . 100
4.138 Parameter ImcrPad . . . 100
4.139 Parameter ImcrMirror . . . 100
4.140 Container Other_Config_List . . . 101
4.141 Parameter OtherName . . . 101
4.142 Parameter OtherNumber . . . 102
4.143 Parameter OthersSiul2Instance . . . 102
4.144 Parameter OtherInput . . . 103
4.145 Parameter OtherMirror . . . 103
4.146 Container CommonPublishedInformation . . . 104
4.147 Parameter ArReleaseMajorVersion . . . 104
4.148 Parameter ArReleaseMinorVersion . . . 105
4.149 Parameter ArReleaseRevisionVersion . . . 105
4.150 Parameter ModuleId . . . 106
4.151 Parameter SwMajorVersion . . . 106
4.152 Parameter SwMinorVersion . . . 107
4.153 Parameter SwPatchVersion . . . 107
4.154 Parameter VendorApiInfix . . . 108
4.155 Parameter VendorId . . . 109

5 Module Index **110**
5.1 Software Specification . . . 110

6 Module Documentation **111**
6.1 AXBS IPV Driver . . . 111
6.1.1 Detailed Description . . . 111
6.1.2 Data Structure Documentation . . . 112
6.1.3 Types Reference . . . 112
6.1.4 Function Reference . . . 113
6.1.5 Variable Documentation . . . 115
6.2 RM Driver . . . 116
6.2.1 Detailed Description . . . 116
6.2.2 Types Reference . . . 117
6.2.3 Function Reference . . . 119
6.2.4 Variable Documentation . . . 130
6.3 Mpu M7 IPV Driver . . . 131
6.3.1 Detailed Description . . . 131

6.3.2 Data Structure Documentation . . . 132
6.3.3 Enum Reference . . . 133
6.3.4 Function Reference . . . 135
6.4 Pflash IPV Driver . . . 142
6.4.1 Detailed Description . . . 142
6.4.2 Data Structure Documentation . . . 142
6.4.3 Enum Reference . . . 143
6.4.4 Function Reference . . . 143
6.4.5 Variable Documentation . . . 145
6.5 Sema42 IPV Driver . . . 146
6.5.1 Detailed Description . . . 146
6.5.2 Enum Reference . . . 147
6.5.3 Function Reference . . . 147
6.6 Virt Wrapper IPV Driver . . . 158
6.6.1 Detailed Description . . . 158
6.6.2 Data Structure Documentation . . . 158
6.6.3 Enum Reference . . . 159
6.6.4 Function Reference . . . 159
6.6.5 Variable Documentation . . . 159
6.7 XBIC IPV Driver . . . 160
6.7.1 Detailed Description . . . 160
6.7.2 Data Structure Documentation . . . 161
6.7.3 Function Reference . . . 162
6.7.4 Variable Documentation . . . 168
6.8 XRDC IPV Driver . . . 169
6.8.1 Detailed Description . . . 169
6.8.2 Data Structure Documentation . . . 170
6.8.3 Enum Reference . . . 173
6.8.4 Function Reference . . . 175
6.8.5 Variable Documentation . . . 180

Chapter 1

Revision History

Revision	Date	Author	Description
1.0	31.03.2022	NXP RTD Team	Prepared for release RTD S32K3 2.0.0

Chapter 2

Introduction

- Supported Derivatives
- Overview
- About This Manual
- Acronyms and Definitions
- Reference List

This User Manual describes Resource Manager for S32K3. Resource Manager driver configuration parameters and deviations from the specification are described in Driver chapter of this document.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32k344_mapbga257
- s32k344_mqfp172
- s32k324_mapbga257
- s32k324_mqfp172
- s32k314_mapbga257
- s32k314_mqfp172
- s32k312_mqfp172
- s32k312_mqfp100
- s32k322_mqfp172
- s32k322_mqfp100
- s32k342_mqfp172
- s32k342_mqfp100
- s32k341_mqfp172
- s32k341_mqfp100

All of the above microcontroller devices are collectively named as S32K3.

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

2.4 Acronyms and Definitions

Term	Definition
API	Application Programming Interface
ASM	Assembler
BSMI	Basic Software Make file Interface
C/CPP	C and C++ Source Code
DEM	Diagnostic Event Manager
DET	Development Error Tracer
ECU	Electronic Control Unit
LSB	Least Signifigant Bit
MCU	Micro Controller Unit
MIDE	Multi Integrated Development Environment
MPU	Memory Protection Unit
MRC	Memory Region Controller
MSB	Most Significant Bit
MPU_M7	Arm Cortex M7 Memory Protection Unit
N/A	Not Applicable
RAM	Random Access Memory
RM	Resource Manager
SEMA42	Semaphores 2
SIU	Systems Integration Unit
SWS	Software Specification
XML	Extensible Markup Language
XRDC	Extended Resource Domain Controller

2.5 Reference List

#	Title	Version
1	S32K3XX Reference Manual	S32K3xx Reference Manual, Rev. 3, 10/2021
2	Datasheet	S32K3xx Data Sheet, Rev. 3 — 10/2021
3	S32K312 Errata	Mask Set Errata for Mask 0P09C, Rev. 10/Jan/2022
4	S32K342 Errata	Mask Set Errata for Mask 0P97C, Rev. 13 Dec 2021
4	S32K344 Errata	Mask Set Errata for Mask 0P55A/1P55A, Rev. 30/Sep/2021
5	Arm Cortex-M7 Devices Generic User Guide (for Mpu M7)	Rev. 15 Nov 2018

Chapter 3

Driver

- Requirements
- Driver Design Summary
- Hardware Resources
- Deviations from Requirements
- Driver Limitations
- Driver usage and configuration tips
- Runtime errors
- Symbolic Names Disclaimer

3.1 Requirements

Resource Manager is a Complex Device Driver (CDD), so there are no AUTOSAR requirements regarding this module.

It has vendor-specific requirements and implementation.

3.2 Driver Design Summary

The RM module provides a way to initialize and control the resource domains allocation and memory protection on the chip with the supported peripherals.

3.3 Hardware Resources

#	Hardware IP	Description
1	AXBS	Crossbar Switch
2	Mpu_M7	Arm Cortex M7 Memory Protection Unit
3	Pflash	Flash Memory Controller
4	Sema42	Semaphores 2
5	Virt_Wrapper	Virtualization Wrapper
6	XBIC	Crossbar Integrity Checker
7	XRDC	Extended Resource Domain Controller

3.4 Deviations from Requirements

Since this is a CDD Module, there are no AUTOSAR requirements for the functionality.

3.5 Driver Limitations

- Functions Xrdc_Ip_GetDomainIDErrorStatus (Ip Layer) and Rm_XrdcGetDomainIDErrorStatus (HLD layer) may return wrong information about attribute error. Example: The program run in unprivileged mode, but attribute error is privileged mode. This is a hardware issue of EATR bit field in DERR_W1 register.
- On S32DS, function Rm_Mpu_M7_GetErrorDetails (Ip layer) and Mpu_M7_Ip_GetErrorDetails (HLD layer) may not work properly because error registers are cleared in the memfault exception handler. The issue may come from PE debugger and it is under investigation

3.6 Driver usage and configuration tips

3.6.1 HLD usage

Prior usage of the RM CDD in an application, the configuration files must be generated with the configurator.

3.6.1.1 API controls

Initialization of all hardware resources is done using Rm_Init. Considering that the funcmion is configuring domains and access to resources it is recommended that it is called before other bus masters are active, or that they are not accesing those resources.

The rest of the API controls directly the allocated hardware resources. Check the references below for more information:

- Rm_XrdcSetProcessID
- Rm_XrdcGetDomainID
- Rm_XrdcGetDomainIDErrorStatus

- Rm_SemaphoreGetStatus
- Rm_SemaphoreLockGate
- Rm_SemaphoreUnlockGate
- Rm_SemaphoreResetGate
- Rm_SemaphoreResetAllGates
- Rm_Mpu_M7_SetRegionConfig
- Rm_Mpu_M7_EnableRegion
- Rm_Mpu_M7_SetAccessRight
- Rm_Mpu_M7_GetErrorDetails
- Rm_Mpu_M7_MulticoreInit
- Rm_XbicEnableMasterFeedbackCheck
- Rm_XbicEnableSlaveEDCCheck
- Rm_XbicFeedbackCheckAndEDCCheckDisable
- Rm_XbicGetErrorStatus
- Rm_XbicErrorInjection
- Rm_XbicErrorInjectionDisable

3.6.2 LLD usage

Prior usage of the LLDs from RM CDD in an application, the configuration files must be generated with the S32CT configurator.

For more information on the LLD API check the references below:

- Mpu M7 IPV Driver
  - Mpu_M7_Ip_Init
  - Mpu_M7_Ip_MulticoreInit
  - Mpu_M7_Ip_SetRegionConfig
  - Mpu_M7_Ip_Deinit
  - Mpu_M7_Ip_EnableRegion
  - Mpu_M7_Ip_SetAccessRight
  - Mpu_M7_Ip_GetErrorDetails
- XRDC IPV Driver
  - Xrdc_Ip_Init
  - Xrdc_Ip_GetDomainID
  - Xrdc_Ip_SetProcessID
  - Xrdc_Ip_GetDomainIDErrorStatus
- Pflash IPV Driver

  - Pflash_Ip_Init
  - Pflash_Ip_SetAccessProtection

- Sema42 IPV Driver
  - Sema42_Ip_Init
  - Sema42_Ip_LockGate
  - Sema42_Ip_UnlockGate
  - Sema42_Ip_ResetGate
  - Sema42_Ip_GetGateLocker
  - Sema42_Ip_IsResetGateStateIdle
  - Sema42_Ip_GetResetGateBusMaster
  - Sema42_Ip_GetResetGateIndex

- AXBS IPV Driver
  - Axbs_Ip_Init
  - Axbs_Ip_DeInit

- XBIC IPV Driver
  - Xbic_Ip_Init
  - Xbic_Ip_EnableMasterFeedbackCheck
  - Xbic_Ip_EnableSlaveEDCCheck
  - Xbic_Ip_FeedbackCheckAndEDCCheckDisable
  - Xbic_Ip_IntegrityCheckerGetErrorStatus
  - Xbic_Ip_IntegrityErrorInjectionConfig
  - Xbic_Ip_IntegrityErrorInjectionDisable

- Virt Wrapper IPV Driver
  - Virt_Wrapper_Ip_Init

3.6.3 Interface configuration

User can choose enable / disable checkbox to use the corresponding Module

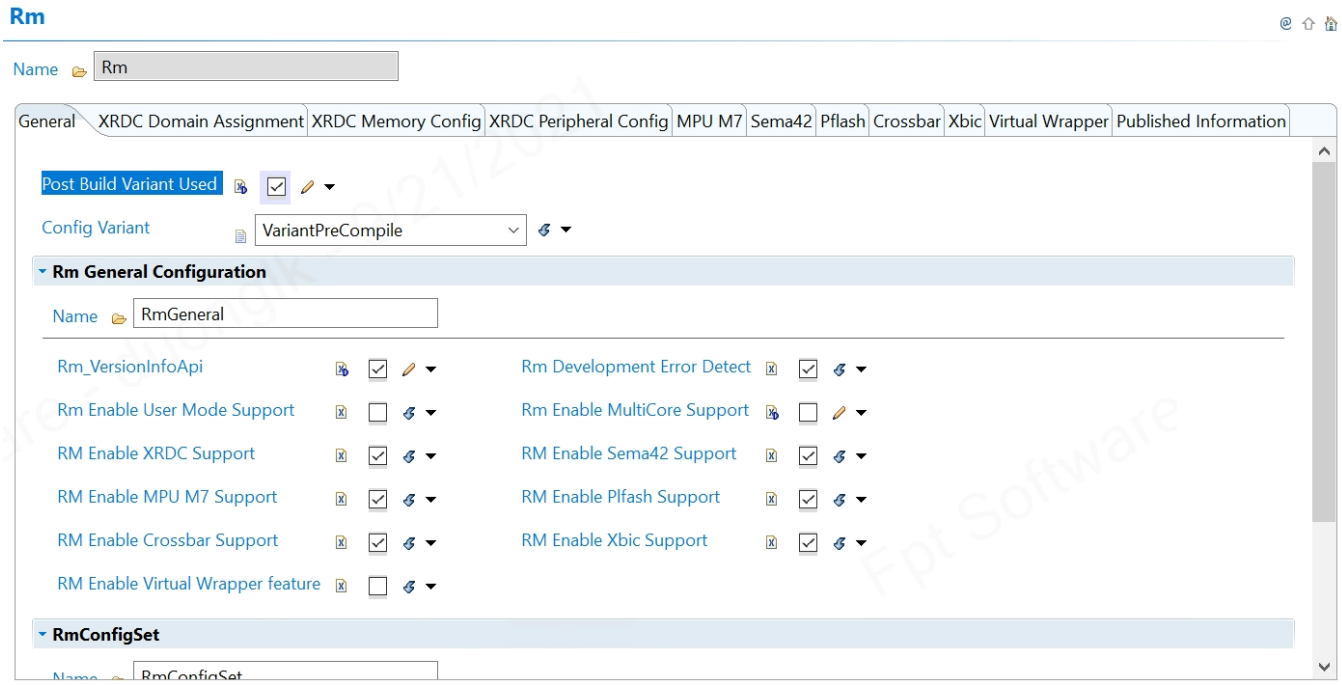


Figure 3.1 RM general config

3.6.4 Multicore initialization

RM is multicore type 3 but MPU can support type 2 multicore following autosar specifications. Because of that it require separated initialization for MPU and the rest of IP driver, If multicore is enable, the following init step should be required:

```
          Core 0                                                    Core 1
step1:  Rm_Init() (for master core only)                    No initilization for Rm_Init()   type 3 multicore
step2:  Do nothing(Mpu M7 Multicore was initialized in RM_Init())  Rm_Mpu_M7_Multicore_Init()      type 2 multicore
```

3.6.5 XRDC configuration

3.6.5.1 XRDC registers lock

When **XRDC Registers Lock** is enable, all Xrdc registers which have LK bit(s) except XRDC_PIDm registers will be locked (read-only) until the next reset.

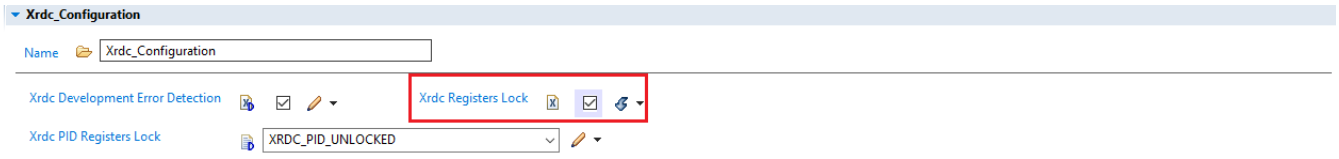


Figure 3.2 XRDC Registers lock

For XRDC PID Registers Lock , Tresos configuration provides users with 3 available options which are listed in figure and table below

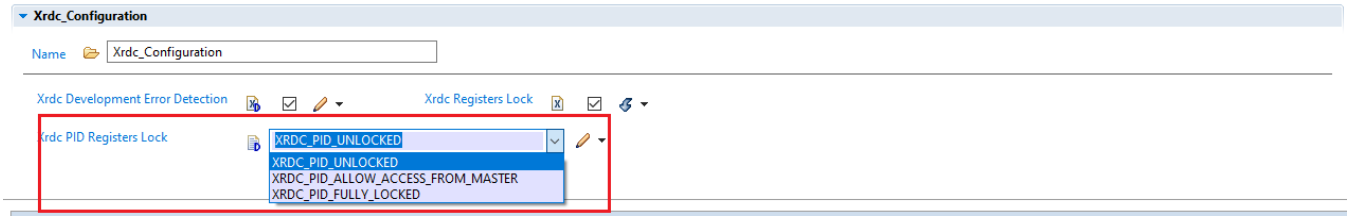


Figure 3.3 XRDC PID Registers lock options

Option	XRDC_PIDm registers status
XRDC_PID_UNLOCKED	Register can be written to by any secure privileged write
XRDC_PID_ALLOW_ACCESS_FROM_MASTER	Register can only be written by a secure privileged write from bus master m
XRDC_PID_FULLY_LOCKED	Register locked (read-only) until the next reset

Note:
-If **XRDC Registers Lock** is enable, XRDC module cannot be disable during runtime. -**XRDC PID Registers Lock** will be aborted unless **XRDC Registers Lock** is enable.

3.6.5.2 XRDC Domain Assignment

Users can create a new domain assignment by click on Add button in configuration field.

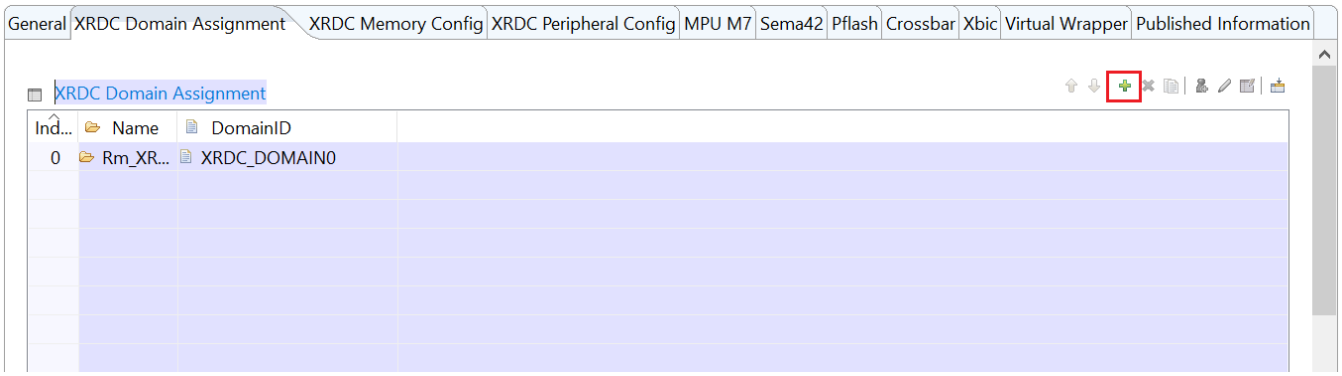


Figure 3.4 XRDC Domain Assignment Configuration

3.6.5.2.1 XRDC Master Domain Assignment

Users can add new elements to choose which masters will be assigned to the corresponding domain ID and configure their assignment attributes.

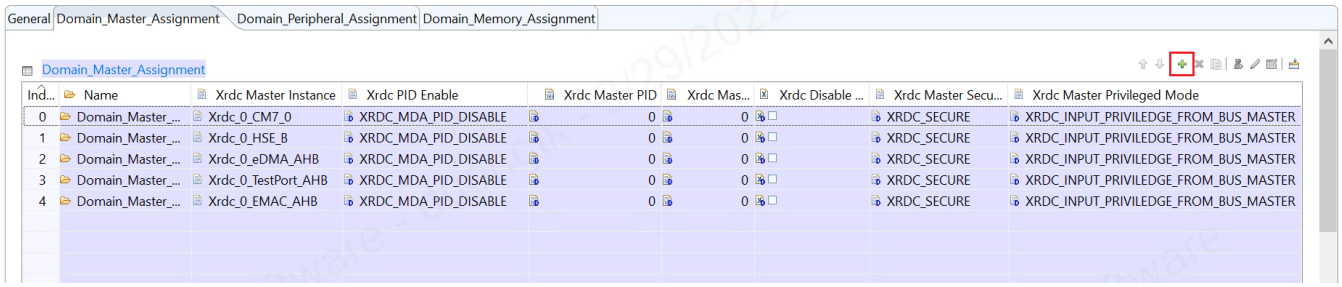


Figure 3.5 XRDC Domain Master Assignment Configuration

For ProcessID evaluation, Tresos configuration provides users with 3 available options which are listed in figure and table below

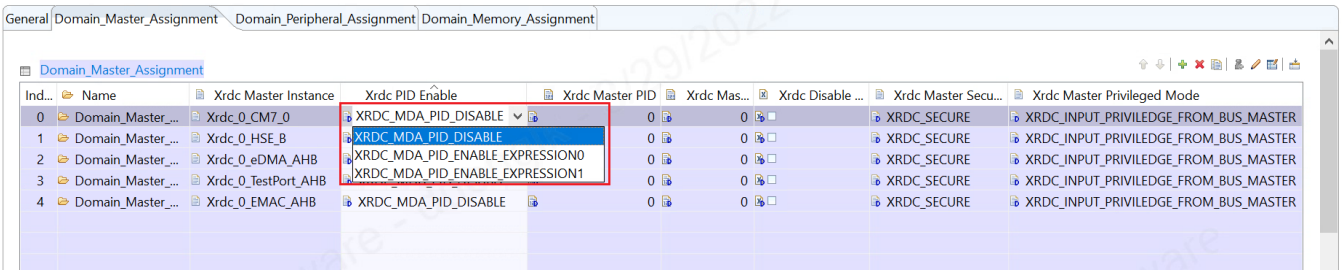


Figure 3.6 ProcessID options

Option	PID evaluation
XRDC_MDA_PID_DISABLE	No process identifier is included in the domain hit evaluation
XRDC_MDA_PID_ENABLE_EXPRESSION0	Domain assignment hit when ((PID & ~PIDM) == (PIDn[PID] &~PIDM))
XRDC_MDA_PID_ENABLE_EXPRESSION1	Domain assignment hit when ~((PID & ~PIDM) == (PIDn[PID] &~PIDM))

3.6.5.2.2 XRDC Peripheral Domain Assignment

Users can add new elements to configure domain access policy for a specific peripheral slot.

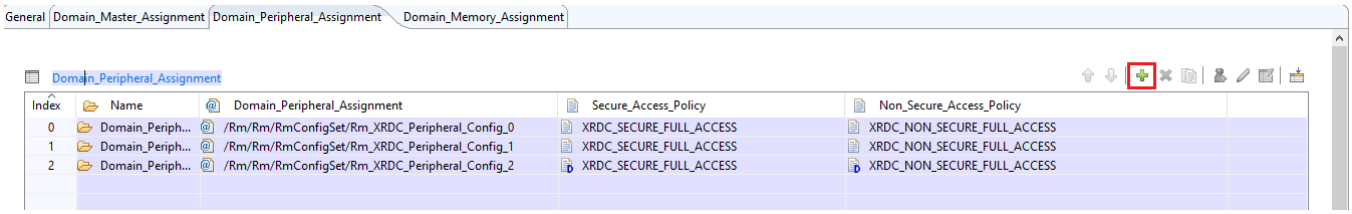


Figure 3.7 XRDC Domain Peripheral Assignment

Selections of **DomainPeripheralAssignment** field refer to XRDC Peripheral Configuration

Combination of **SecurityAccessPolicy** and **NonSecurityAccessPolicy** field defines final access policy for the peripheral slot (found in ***Access policy look up table*** shown below)

3.6.5.2.3 XRDC Memory Domain Assignment

Users can add new elements to configure domain access policy for a specific memory region.

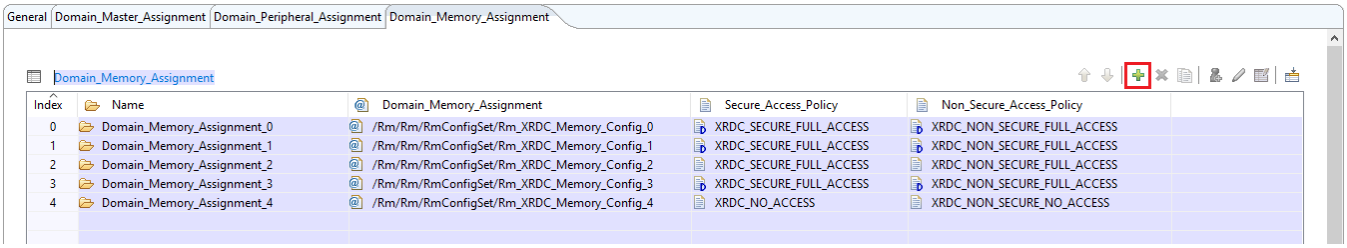


Figure 3.8 XRDC Domain Memory Assignment

Selections of **DomainMemoryAssignment** field refer to XRDC Memory Configuration

Combination of **SecurityAccessPolicy** and **NonSecurityAccessPolicy** field defines final access policy for the memory region (found in ***Access policy look up table*** shown below)

Configuration Fields		Allowable accesses			
Secure_Access_Policy	Non_Secure_Access_Policy	Secure Privileged	Secure User	NonSecure Privileged	NonSecure User
XRDC_FULL_ACCESS	XRDC_FULL_ACCESS	R,W	R,W	R,W	R,W
XRDC_FULL_ACCESS	XRDC_SUPERVISOR_ACCESS_ONLY	R,W	R,W	R,W	None
XRDC_FULL_ACCESS	XRDC_READ_ONLY	R,W	R,W	R	R
XRDC_FULL_ACCESS	XRDC_SUPERVISOR_READ_ONLY	R,W	R,W	R	None
XRDC_FULL_ACCESS	XRDC_NO_ACCESS	R,W	R,W	None	None
XRDC_SUPERVISOR_ACCESS_ONLY	NA	R,W	None	None	None
XRDC_READ_ONLY	NA	R	R	None	None
XRDC_NO_ACCESS	NA	None	None	None	None

Figure 3.9 Access policy look up table

3.6.5.3 XRDC Peripheral Configuration

Users can add new elements to configure attributes of peripheral slots

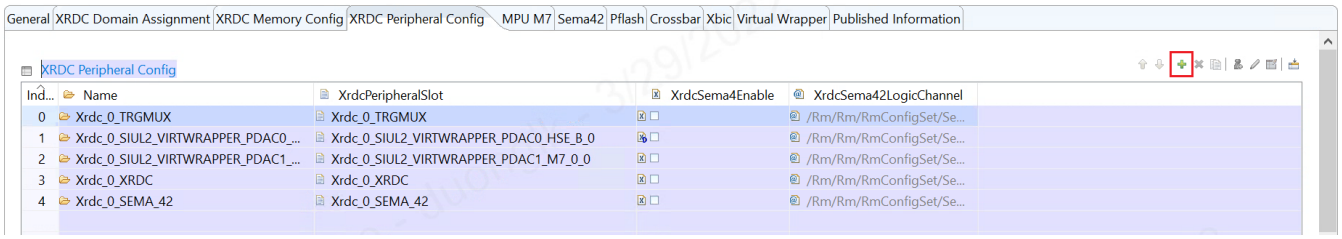


Figure 3.10 XRDC Domain Peripheral Configuration

Note:
-**XrdcSema42LogicChannel** will be aborted unless **XrdcSema4Enable** is checked.
-**XrdcSema42LogicChannel** can be mapped to logical channels mentioned in Sema42 logic channel configuration

3.6.5.4 XRDC Memory Configuration

Users can add new elements to configure attributes of memory regions.

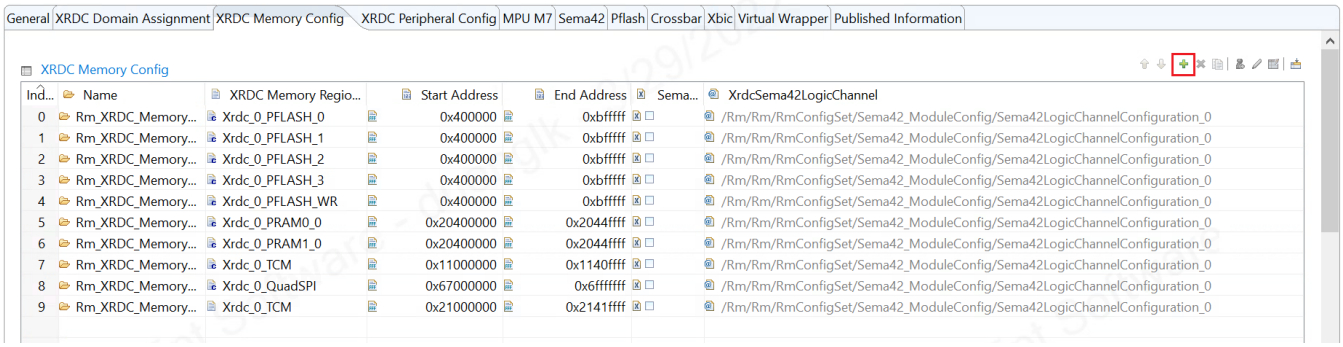


Figure 3.11 XRDC Domain Memory Configuration

Note:
-**XrdcSema42LogicChannel** will be aborted unless **XrdcSema4Enable** is checked.
-**XrdcSema42LogicChannel** can be mapped to logical channels mentioned in Sema42 logic channel configuration
-Each memory region need to be protected by an appropriate MRC instance, details are mentioned in the Reference manual

3.6.6 Sema42 logic channel configuration

Users can use logic channel instead of hardware instance and hardware channel. The logic channel is stand for both hardware instance and hardware channel which is configured in configration application.

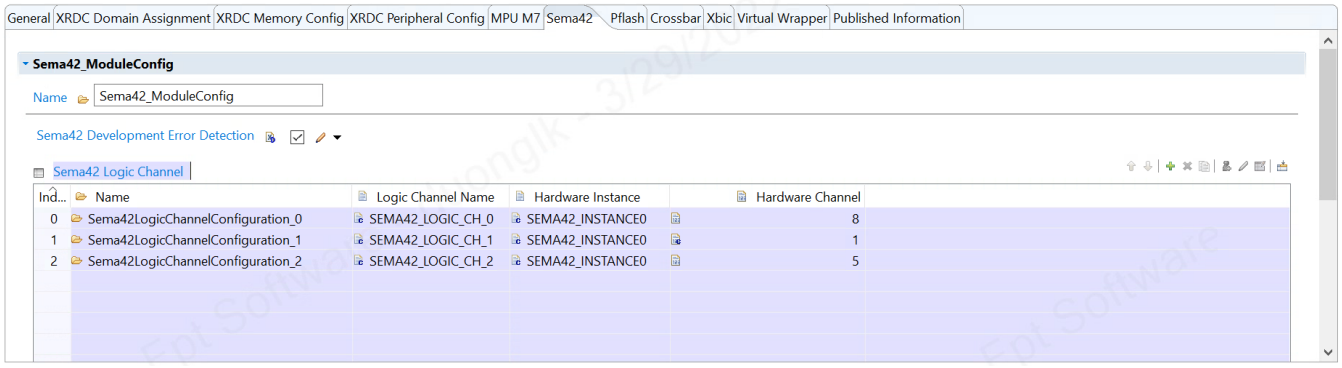


Figure 3.12 Sema42 LogicChannel Configuration

Example: In the picture, there are 3 logic channel:

- Logic channel 1 is equal to Sema42 instance 0 gate 8.
- Logic channel 2 is equal to Sema42 instance 0 gate 1.
- Logic channel 3 is equal to Sema42 instance 0 gate 5.

Note: In Rm_Semaphore API, we should use the name which is defined in column **Logic Channel Name** in configuration application with the prefix "RM_" (Because it is a macro so all characters are uppercase). We can find out them in file "CDD_Rm_Ipw_Cfg.h" with identify "RM_<Logic Channel Name>". As in picture, if users want to invoke Sema42 instance 0 gate 5, the argument they must pass into Rm_Semaphore API is RM_SEMA42_L↩ OGIC_CH_2.

3.6.7 MPU M7 configuration

The MPU divides the memory map into a number of regions, and defines the location, size, access permissions, and memory attributes of each region. Details refer to "ARMv7-M Architecture Reference Manual" documentation of the ARM. (Chapter B3.5 Protected Memory System Architecture, PMSAv7)

A Mpu M7 Region Configuration Interface as shown below:

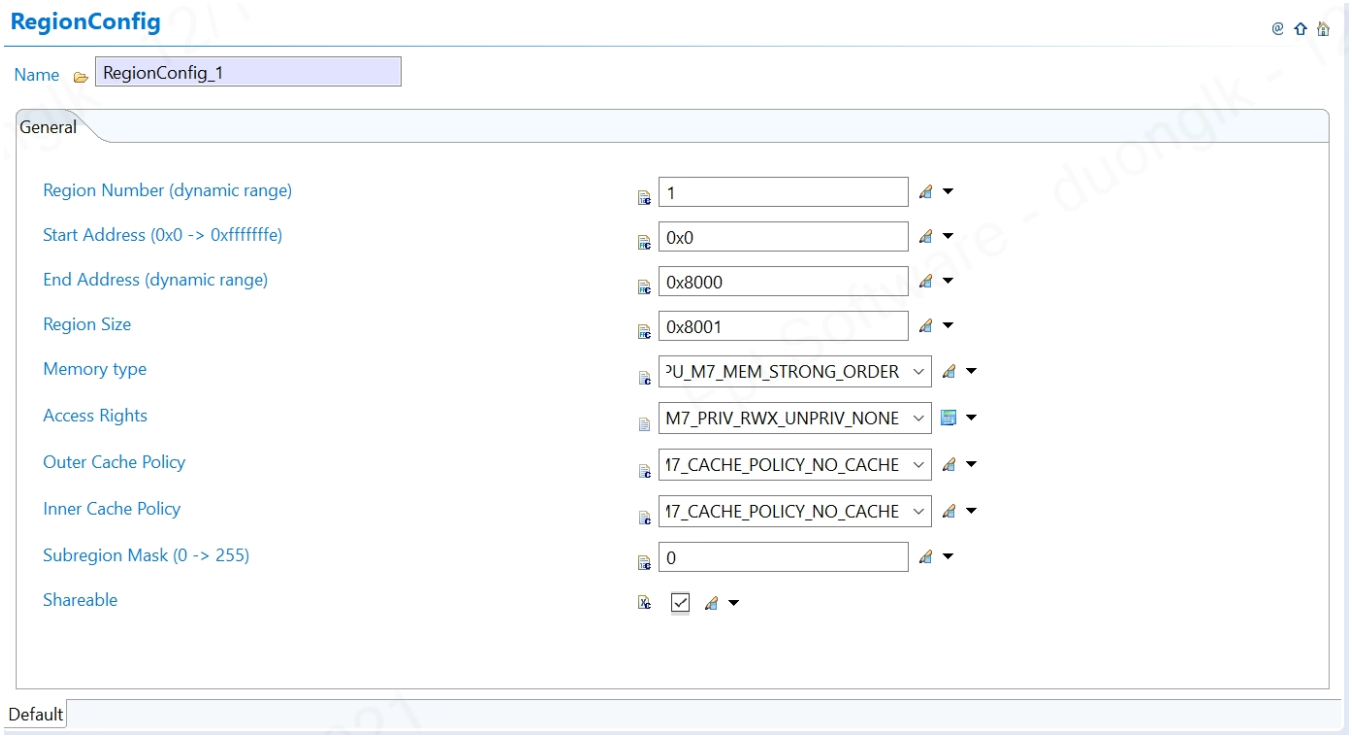


Figure 3.13 Mpu M7 Region Configuration Interface.

The **Memory Type** node allows to select the attribute for the memory region shown in the picture below.

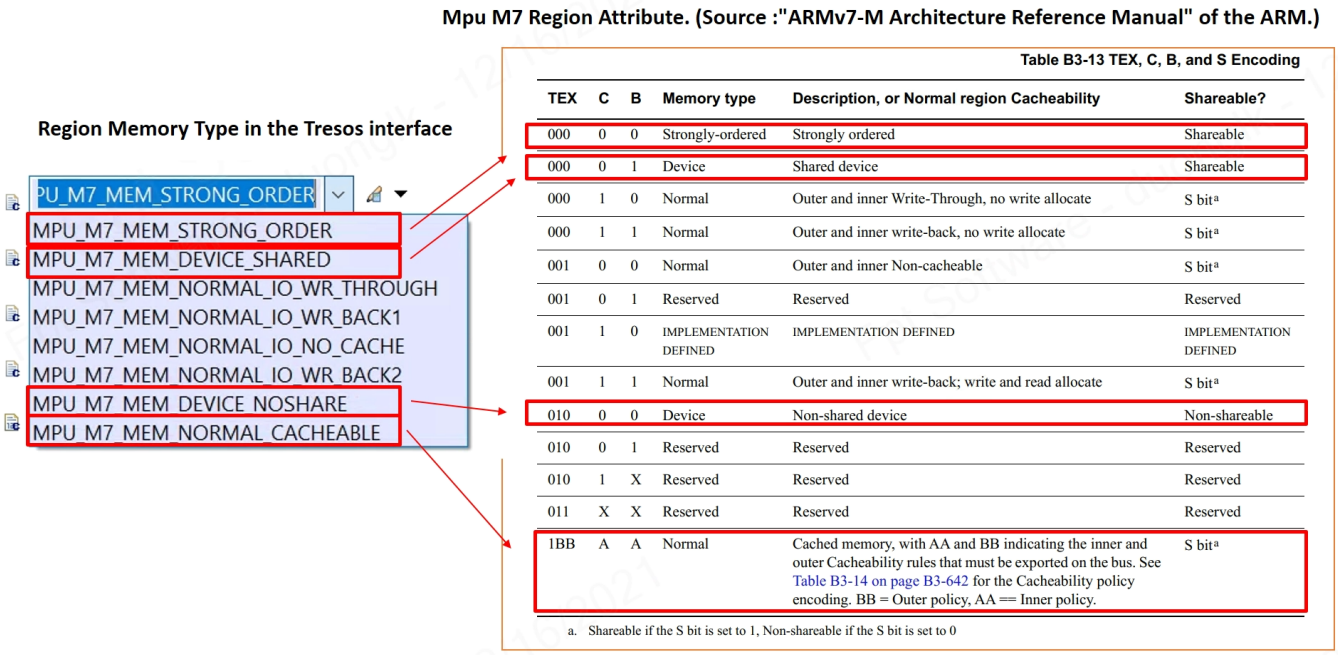


Figure 3.14 Mpu M7 Region Memory Type.

Note:
-Only if the configuration **Memory type** is "MPU_M7_MEM_NORMAL_CACHEABLE", "Outer Cache Policy"/"Inner Cache Policy" makes sense.
-When configuration **Memory type** is "MPU_M7_MEM_STRONG_ORDER", "MPU_M7_MEM_DEVICE_↩ SHARED" , means region is always Shareable, the **Shareable** node value has no meaning.
-When configuration **Memory type** is "MPU_M7_MEM_DEVICE_NOSHARE" , means region is always Non-↩ Shareable, the **Shareable** node value has no meaning.

3.6.8 Errata for MPU M7 : [Core] Speculative accesses might be performed to memory unmapped in MPU.

Cortex-M7 can incorrectly perform speculative accesses to unmapped regions. Instead of leaving memory unmapped, the RM driver recommends the user to use MPU region 0 to cover all memory (0x0-0xffffffff) and make this region execute-never and inaccessible. In tresos interface, default region 0 should be configured as shown below:

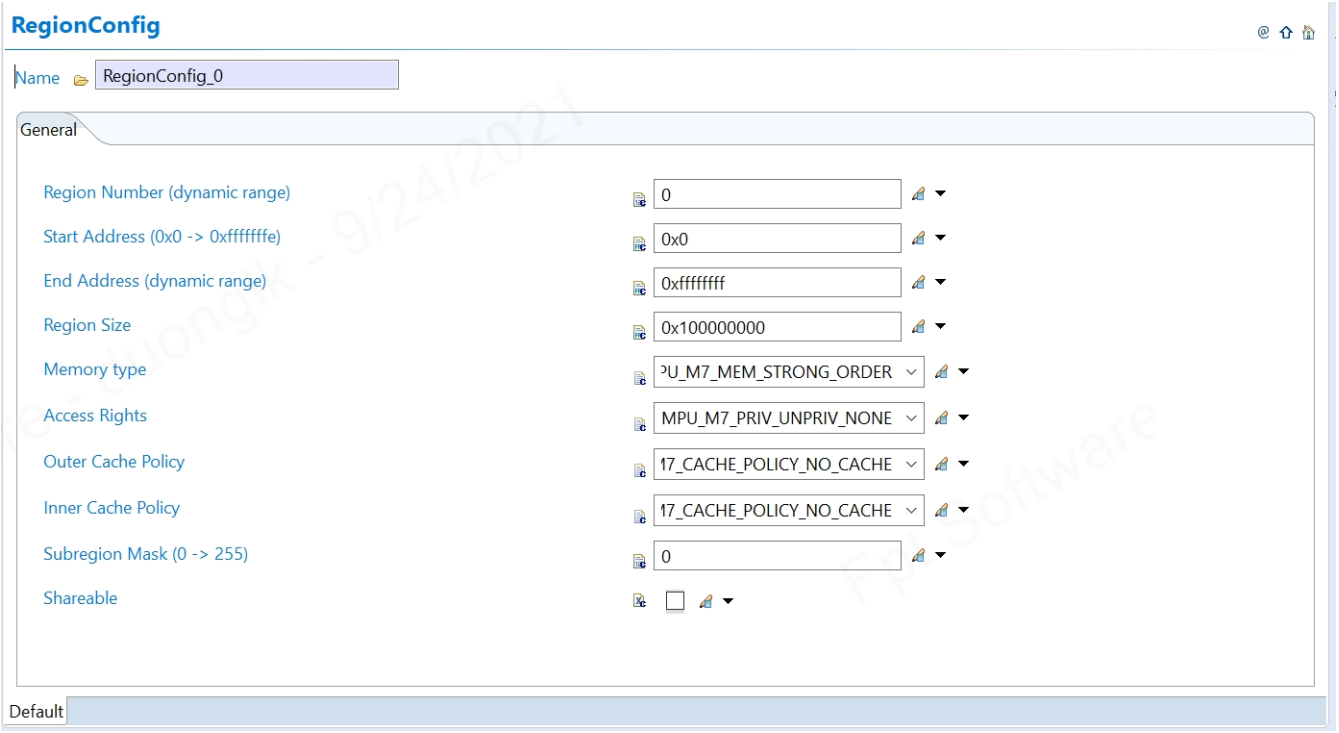


Figure 3.15 Default region 0

Note that the MPU supports addressing hitting in multiple regions with the highest numbered region taking priority. Therefore, use of MPU region 0 in this way does not affect the existing organization and use of MPU regions

3.7 Runtime errors

The driver generates the following DET errors at runtime.

Function	Error Code	Condition triggering the error
Rm_Init()	RM_E_UNINIT_U8	API service called without module initialization.
Rm_Init()	RM_E_ALREADY_INITIALIZED_U8	Module is already initialized
Rm_Init()	RM_E_INIT_FAILED_U8	In precompile mode, specifies that a non-null parameter was used for init. In other cases, a null pointer was used for initialization
Rm_GetVersionInfo()	RM_E_PARAM_POINTER	Input pointer parameter is invalid
Rm_SemaphoreGetStatus(), Rm_SemaphoreLockGate(), Rm_SemaphoreUnlockGate(), Rm_SemaphoreResetGate()	RM_E_INVALID_SEMA4_CHANNEL_U8	Semaphore channel is invalid
Rm_SemaphoreLockGate()	RM_E_ALREADY_LOCKED_U8	Semaphore is already locked

Function	Error Code	Condition triggering the error
Rm_SemaphoreLockGate(), Rm_SemaphoreUnlockGate()	RM_E_LOCKED_BY_OTHER_CORE_U8	Semaphore is locked by another core
Rm_SemaphoreLockGate()	RM_E_ALREADY_UNLOCKED_U8	Semaphore is already unlocked

3.8 Symbolic Names Disclaimer

All containers having symbolicNameValue set to TRUE in the AUTOSAR schema will generate defines like:

#define <Mip>Conf_<Container_ShortName>_<Container_ID>

For this reason it is forbidden to duplicate the names of such containers across the RTD configurations or to use names that may trigger other compile issues (e.g. match existing `#ifdefs` arguments).

Chapter 4

Tresos Configuration Plug-in

This chapter describes the Tresos configuration plug-in for the driver. All the parameters are described below.

- Module Rm
  - Container RmGeneral
    - Parameter Rm_VersionInfoApi
    - Parameter RmDevErrorDetect
    - Parameter RmEnableUserModeSupport
    - Parameter RmEnableMultiCore
    - Parameter RmEnableXRDCSupport
    - Parameter RmEnableSema42Support
    - Parameter RmMpuM7Configurable
    - Parameter RmPflashConfigurable
    - Parameter RmCrossbarConfigurable
    - Parameter RmXbicConfigurable
    - Parameter RmVirtWrapperConfigurable
  - Container RmConfigSet
    - Container Xrdc_Configuration
      - Parameter XrdcDevErrorDetect
      - Parameter XrdcRegistersLock
      - Parameter XrdcPIDRegisterLock
    - Container Rm_XRDC_Domain_Assignment
      - Parameter DomainID
      - Container Domain_Master_Assignment
      - Parameter XrdcMasterInstance
      - Parameter XrdcPIDEnable
      - Parameter XrdcMasterPID
      - Parameter XrdcMasterPIDMask
      - Parameter XrdcDIDBDisable
      - Parameter XrdcMasterMode
      - Parameter XrdcMasterPriviledgeMode
      - Container Domain_Peripheral_Assignment
      - Parameter Secure_Access_Policy

- Parameter Non_Secure_Access_Policy
- Reference Domain_Peripheral_Assignment
- Container Domain_Memory_Assignment
- Parameter Secure_Access_Policy
- Parameter Non_Secure_Access_Policy
- Reference Domain_Memory_Assignment

* Container Rm_XRDC_Memory_Config
  - Parameter XrdcMrcInstance
  - Parameter XrdcStartAddress
  - Parameter XrdcEndAddress
  - Parameter XrdcSema4Enable
  - Reference XrdcSema42LogicChannel
* Container Rm_XRDC_Peripheral_Config
  - Parameter XrdcPeripheralSlot
  - Parameter XrdcSema4Enable
  - Reference XrdcSema42LogicChannel
* Container MPU_M7_Configuration
  - Parameter MpuDevErrorDetect
  - Parameter ClearCacheEnable
  - Container MPU_M7_ModuleConfig
  - Parameter DefaultMapEnable
  - Parameter RunInHFNMIEnable
  - Parameter MemManageInterruptEnable
  - Parameter PhysicalCoreID
  - Container RegionConfig
  - Parameter RegionNumber
  - Parameter StartAddress
  - Parameter EndAddress
  - Parameter RegionSize
  - Parameter MemoryType
  - Parameter AccessRights
  - Parameter OuterCachePolicy
  - Parameter InnerCachePolicy
  - Parameter SubregionMask
  - Parameter Shareable

- Container Sema42_ModuleConfig
  - Parameter Sema42DevErrorDetect
  - Container Sema42LogicChannelConfiguration
  - Parameter Sema42LogicChannel_LogicName
  - Parameter Sema42HardwareInstance
  - Parameter Sema42HardwareChannel
- Container Pflash_Configuration
  - Parameter PflashDevErrorDetect
  - Container PflashMasterProtection
  - Parameter PflashControler
  - Parameter PflashMAP
  - Parameter PflashMasterAccess
- Container Crossbar_Configuration
  - Parameter CrossbarDevErrorDetect
  - Container RmCrossbarInstance
  - Parameter RmCrossbarHwInstance
  - Container RmCrossbarHwSlavePort
  - Parameter RmSlavePortNumber
  - Parameter RmCrossbarPrioMaster0
  - Parameter RmCrossbarPrioMaster1
  - Parameter RmCrossbarPrioMaster2
  - Parameter RmCrossbarPrioMaster3
  - Parameter RmCrossbarPrioMaster4
  - Parameter RmCrossbarPrioMaster5
  - Parameter RmCrossbarPrioMaster6
  - Parameter RmCrossbarPrioMaster7
  - Parameter RmCrossbarEnableLock
  - Parameter RmCrossbarHaltLowPrio
  - Parameter RmCrossbarEnablePrioElevM0
  - Parameter RmCrossbarEnablePrioElevM1
  - Parameter RmCrossbarEnablePrioElevM2
  - Parameter RmCrossbarEnablePrioElevM3
  - Parameter RmCrossbarEnablePrioElevM4
  - Parameter RmCrossbarEnablePrioElevM5
  - Parameter RmCrossbarEnablePrioElevM6
  - Parameter RmCrossbarEnablePrioElevM7
  - Parameter RmCrossbarEnableFixedPrio
  - Parameter RmCrossbarParkingControl
  - Parameter RmCrossbarParkField
- Container Xbic_Configuration
  - Parameter XbicDevErrorDetect
  - Container XbicCheckingControl
  - Parameter XbicInstance
  - Parameter SE0
  - Parameter SE1
  - Parameter SE2
  - Parameter SE3

- Parameter SE4
- Parameter SE5
- Parameter SE6
- Parameter SE7
- Parameter ME0
- Parameter ME1
- Parameter ME2
- Parameter ME3
- Parameter ME4
- Parameter ME5
- Parameter ME6
- Parameter ME7

* Container Virt_Wrapper_Configuration
  - Parameter VirtWrapperDevErrorDetect
  - Container Mscr_Config_List
  - Parameter MscrName
  - Parameter MscrNumber
  - Parameter MscrSiul2Instance
  - Parameter MscrPinMux
  - Parameter MscrMirror
  - Container Imcr_Config_List
  - Parameter ImcrName
  - Parameter ImcrNumber
  - Parameter ImcrSiul2Instance
  - Parameter ImcrInput
  - Parameter ImcrPad
  - Parameter ImcrMirror
  - Container Other_Config_List
  - Parameter OtherName
  - Parameter OtherNumber
  - Parameter OthersSiul2Instance
  - Parameter OtherInput
  - Parameter OtherMirror

– Container CommonPublishedInformation

* Parameter ArReleaseMajorVersion
* Parameter ArReleaseMinorVersion
* Parameter ArReleaseRevisionVersion
* Parameter ModuleId
* Parameter SwMajorVersion
* Parameter SwMinorVersion
* Parameter SwPatchVersion
* Parameter VendorApiInfix
* Parameter VendorId

4.1 Module Rm

Vendor specific: Configuration of the Rm (Resource Manager) module.

Included containers:

- RmGeneral
- RmConfigSet
- CommonPublishedInformation

Property	Value
type	ECUC-MODULE-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-POST-BUILD, VARIANT-PRE-COMPILE

4.2 Container RmGeneral

Vendor specific: Configuration of general Rm parameters.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.3 Parameter Rm_VersionInfoApi

Vendor specific: Enables/Disables the get version info API function

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.4 Parameter RmDevErrorDetect

Vendor specific:

Switches the Development Error Detection and Notification on or off.

true: Enabled.

false: Disabled.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.5 Parameter RmEnableUserModeSupport

When this parameter is enabled, the RM module will adapt to run from User Mode, with the following measures:
b) using 'call trusted function' stubs for all internal function calls that access registers requiring supervisor mode.
for more information, please see chapter 5.7 User Mode Support in IM
Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.6 Parameter RmEnableMultiCore

When this parameter is enabled, the RM module will adapt to run with Multicore:

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.7 Parameter RmEnableXRDCSupport

When this parameter is enabled, the Xrdc APIs are enabled to the user

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.8 Parameter RmEnableSema42Support

When this parameter is enabled, the Sema42 APIs are enabled to the user

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.9 Parameter RmMpuM7Configurable

RmMpuM7Configurable

Check this in order to be able to use the ARM Cortex M7 MPU.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.10 Parameter RmPflashConfigurable

RmPflashConfigurable

Check this in order to be able to use the Master protection.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.11 Parameter RmCrossbarConfigurable

RmCrossbarConfigurable

Check this in order to be able to use the crossbar configuration feature.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.12 Parameter RmXbicConfigurable

RmXbicConfigurable

Check this in order to be able to use XBIC api sets to verify the integrity of crossbar transfers.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.13 Parameter RmVirtWrapperConfigurable

RmVirtWrapperConfigurable

Check this in order to be able Virtual wrapper feature. This feature need to enable along side with port virtual wrapper feature to ensure collaboration set of pads protection. When virtual wrapper is enable RM driver will automatically ajust configuration for hw protection.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.14 Container RmConfigSet

Vendor specific: This container is the base for a multiple configuration set

Included subcontainers:

- Xrdc_Configuration
- Rm_XRDC_Domain_Assignment
- Rm_XRDC_Memory_Config
- Rm_XRDC_Peripheral_Config
- MPU_M7_Configuration
- Sema42_ModuleConfig
- Pflash_Configuration
- Crossbar_Configuration
- Xbic_Configuration
- Virt_Wrapper_Configuration

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.15 Container Xrdc_Configuration

Configuration for the Xrdc.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.16 Parameter XrdcDevErrorDetect

XrdcDevErrorDetect

Switches the Development Error Detection and Notification ON or OFF.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.17 Parameter XrdcRegistersLock

Lock XRDC registers (except PIDm registers) in order to prevent them from being unintentionally modified during Runtime.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.18 Parameter XrdcPIDRegisterLock

Select lock mode for PIDm registers

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	XRDC_PID_UNLOCKED
literals	['XRDC_PID_UNLOCKED', 'XRDC_PID_ALLOW_ACCESS_FROM_M↩ ASTER', 'XRDC_PID_FULLY_LOCKED']

4.19 Container Rm_XRDC_Domain_Assignment

All data need to configure one Domain.

Included subcontainers:

- Domain_Master_Assignment
- Domain_Peripheral_Assignment
- Domain_Memory_Assignment

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.20 Parameter DomainID

Vendor specific:

Id for the current Domain.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	XRDC_DOMAIN0
literals	['XRDC_DOMAIN0', 'XRDC_DOMAIN1', 'XRDC_DOMAIN2']

4.21 Container Domain_Master_Assignment

Vendor specific:

All data need to configure one Master.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.22 Parameter XrdcMasterInstance

Select the master of XRDC

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Xrdc_0_CM7_0
literals	['Xrdc_0_CM7_0', 'Xrdc_0_eDMA_AHB', 'Xrdc_0_TestPort_AHB', 'Xrdc↩ _0_HSE_B', 'Xrdc_0_reserved', 'Xrdc_0_EMAC_AHB']

4.23 Parameter XrdcPIDEnable

PID mode enable - only valid if the master instance is core type

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	XRDC_MDA_PID_DISABLE
literals	['XRDC_MDA_PID_DISABLE', 'XRDC_MDA_PID_ENABLE_EXPRESSION0', 'XRDC_MDA_PID_ENABLE_EXPRESSION1']

4.24 Parameter XrdcMasterPID

Process Identifier. This field only has meaning if the core is core master type and PID mode enable

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	63
min	0

4.25 Parameter XrdcMasterPIDMask

Process Identifier Mask. This field only has meaning if the core is core master type and PID mode enable

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD

Property	Value
defaultValue	0
max	63
min	0

4.26 Parameter XrdcDIDBDisable

Disabled DID bypass. This field only has meaning if the core is non-core master type

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.27 Parameter XrdcMasterMode

Specifies the secure attribute. This field only has meaning if the core is non-core master type

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	XRDC_SECURE
literals	['XRDC_SECURE', 'XRDC_NON_SECURE', 'XRDC_INPUT_FROM_BUS_MASTER']

4.28 Parameter XrdcMasterPriviledgeMode

Specifies the privileged (supervisor/user) attribute. This field only has meaning if the core is non-core master type

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	XRDC_INPUT_PRIVILEDGE_FROM_BUS_MASTER
literals	['XRDC_USER_MODE', 'XRDC_PRIVILEDGE_MODE', 'XRDC_INPUT↩ _PRIVILEDGE_FROM_BUS_MASTER']

4.29 Container Domain_Peripheral_Assignment

Vendor specific:

All data need to configure Peripheral for one Domain.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.30 Parameter Secure_Access_Policy

Vendor specific:

Mode access to policy of mode Secure.

Access policy look up table:

Configuration Fields

Allowable accesses

Secure_Access_Policy

Non_Secure_Access_Policy

Secure Privileged

Secure User

NonSecure Privileged

NonSecure User

XRDC_FULL_ACCESS

XRDC_FULL_ACCESS

R,W

R,W

R,W

R,W

XRDC_FULL_ACCESS

XRDC_SUPERVISOR_ACCESS_ONLY

R,W

R,W

R,W

None

XRDC_FULL_ACCESS

XRDC_READ_ONLY

R,W

R,W

R

R

XRDC_FULL_ACCESS

XRDC_SUPERVISOR_READ_ONLY

R,W

R,W

R

None

XRDC_FULL_ACCESS

XRDC_NO_ACCESS

R,W

R,W

None

None

XRDC_SUPERVISOR_ACCESS_ONLY

NA

R,W

None

None

None

XRDC_READ_ONLY

NA

R

R

None

None

XRDC_NO_ACCESS

NA

None

None

None

None

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	XRDC_FULL_ACCESS
literals	['XRDC_FULL_ACCESS', 'XRDC_READ_ONLY', 'XRDC_SUPERVISOR↩ _ACCESS_ONLY', 'XRDC_NO_ACCESS']

4.31 Parameter Non_Secure_Access_Policy

Vendor specific:

Mode access to policy of mode Non_Secure.

Access policy look up table:

Configuration Fields

Allowable accesses

Secure_Access_Policy

Non_Secure_Access_Policy

Secure Privileged

Secure User

NonSecure Privileged

NonSecure User

XRDC_FULL_ACCESS

XRDC_FULL_ACCESS

R,W

R,W

R,W

R,W

XRDC_FULL_ACCESS

XRDC_SUPERVISOR_ACCESS_ONLY

R,W

R,W

R,W

None

XRDC_FULL_ACCESS

XRDC_READ_ONLY

R,W

R,W

R

R

XRDC_FULL_ACCESS

XRDC_SUPERVISOR_READ_ONLY

R,W

R,W

R

None

XRDC_FULL_ACCESS

XRDC_NO_ACCESS

R,W

R,W

None

None

XRDC_SUPERVISOR_ACCESS_ONLY

NA

R,W

None

None

None

XRDC_READ_ONLY

NA

R

R

None

None

XRDC_NO_ACCESS

NA

None

None

None

None

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	XRDC_FULL_ACCESS
literals	['XRDC_FULL_ACCESS', 'XRDC_SUPERVISOR_ACCESS_ONLY', 'XR↩ DC_READ_ONLY', 'XRDC_SUPERVISOR_READ_ONLY', 'XRDC_NO_↩ ACCESS']

4.32 Reference Domain_Peripheral_Assignment

Vendor specific:

Reference to Peripheral config.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/TS_T40D34M20I0R0/Rm/RmConfigSet/Rm_XRDC_Peripheral_Config

4.33 Container Domain_Memory_Assignment

Vendor specific:

All data need to configure Memory for one Domain.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.34 Parameter Secure_Access_Policy

Vendor specific:

Mode access to policy of mode Secure.

Access policy look up table:

Configuration Fields

Allowable accesses

Secure_Access_Policy

Non_Secure_Access_Policy

Secure Privileged

Secure User

NonSecure Privileged

NonSecure User

XRDC_FULL_ACCESS

XRDC_FULL_ACCESS

R,W

R,W

R,W

R,W

XRDC_FULL_ACCESS

XRDC_SUPERVISOR_ACCESS_ONLY

R,W

R,W

R,W

None

XRDC_FULL_ACCESS

XRDC_READ_ONLY

R,W

R,W

R

R

XRDC_FULL_ACCESS

XRDC_SUPERVISOR_READ_ONLY

R,W

R,W

R

None

XRDC_FULL_ACCESS

XRDC_NO_ACCESS

R,W

R,W

None

None

XRDC_SUPERVISOR_ACCESS_ONLY

NA

R,W

None

None

None

XRDC_READ_ONLY

NA

R

R

None

None

XRDC_NO_ACCESS

NA

None

None

None

None

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	XRDC_FULL_ACCESS
literals	['XRDC_FULL_ACCESS', 'XRDC_READ_ONLY', 'XRDC_SUPERVISOR↩ _ACCESS_ONLY', 'XRDC_NO_ACCESS']

4.35 Parameter Non_Secure_Access_Policy

Vendor specific:

Mode access to policy of mode Non_Secure.

Access policy look up table:

Configuration Fields

Allowable accesses

Secure_Access_Policy

Non_Secure_Access_Policy

Secure Privileged

Secure User

NonSecure Privileged

NonSecure User

XRDC_FULL_ACCESS

XRDC_FULL_ACCESS

R,W

R,W

R,W

R,W

XRDC_FULL_ACCESS

XRDC_SUPERVISOR_ACCESS_ONLY

R,W

R,W

R,W

None

XRDC_FULL_ACCESS

XRDC_READ_ONLY

R,W

R,W

R

R

XRDC_FULL_ACCESS

XRDC_SUPERVISOR_READ_ONLY

R,W

R,W

R

None

XRDC_FULL_ACCESS

XRDC_NO_ACCESS

R,W

R,W

None

None

XRDC_SUPERVISOR_ACCESS_ONLY

NA

R,W

None

None

None

XRDC_READ_ONLY

NA

R

R

None

None

XRDC_NO_ACCESS

NA

None

None

None

None

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	XRDC_FULL_ACCESS
literals	['XRDC_FULL_ACCESS', 'XRDC_SUPERVISOR_ACCESS_ONLY', 'XR↩DC_READ_ONLY', 'XRDC_SUPERVISOR_READ_ONLY', 'XRDC_NO_↩ACCESS']

4.36 Reference Domain_Memory_Assignment

Choose Memory configuration

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/TS_T40D34M20I0R0/Rm/RmConfigSet/Rm_XRDC_Memory_Config

4.37 Container Rm_XRDC_Memory_Config

Vendor specific:

All data need to configure Memory use for Domain.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.38 Parameter XrdcMrcInstance

Select Memory region controller will be used to control the address range below

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Xrdc_0_PFLASH_0
literals	['Xrdc_0_PFLASH_0', 'Xrdc_0_PFLASH_1', 'Xrdc_0_PFLASH_2', 'Xrdc↩ _0_PFLASH_3', 'Xrdc_0_PFLASH_WR', 'Xrdc_0_PRAM0_0', 'Xrdc_0_↩ PRAM1_0', 'Xrdc_0_TCM', 'Xrdc_0_QuadSPI']

4.39 Parameter XrdcStartAddress

Start address of Memory region

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	412316860416
min	0

4.40 Parameter XrdcEndAddress

End address of Memory region

Property	Value
type	ECUC-INTEGER-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	412316860416
min	0

4.41 Parameter XrdcSema4Enable

Enable Semaphore would require the domain must own the gate to have access right for this memory range

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.42 Reference XrdcSema42LogicChannel

Select XrdcSema42LogicChannel to select specified Sema42 Logic Channel

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/TS_T40D34M20I0R0/Rm/RmConfigSet/Sema42_ModuleConfig/Sema42↩ LogicChannelConfiguration

4.43 Container Rm_XRDC_Peripheral_Config

Vendor specific:

All data need to configure Peripheral for Domain.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.44 Parameter XrdcPeripheralSlot

Peripheral slot number of a block of peripherals. Checking Reference manual for chip specific

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Xrdc_0_TRGMUX
literals	['Xrdc_0_TRGMUX', 'Xrdc_0_BCTU', 'Xrdc_0_EMIOS_0', 'Xrdc_0_EMIOS_1', 'Xrdc_0_EMIOS_2', 'Xrdc_0_LCU_0', 'Xrdc_0_LCU_1', 'Xrdc_0_ADC_0', 'Xrdc_0_ADC_1', 'Xrdc_0_ADC_2', 'Xrdc_0_PIT_0', 'Xrdc_0_PIT_1', 'Xrdc_0_AXBS', 'Xrdc_0_System_XBIC', 'Xrdc_0_Periph_XBIC', 'Xrdc_0_EDMA', 'Xrdc_0_EDMA_TCD_0', 'Xrdc_0_EDMA_TCD_1', 'Xrdc_0_EDMA_TCD_2', 'Xrdc_0_EDMA_TCD_3', 'Xrdc_0_EDMA_TCD_4', 'Xrdc_0_EDMA_TCD_5', 'Xrdc_0_EDMA_TCD_6', 'Xrdc_0_EDMA_TCD_7', 'Xrdc_0_EDMA_TCD_8', 'Xrdc_0_EDMA_TCD_9', 'Xrdc_0_EDMA_TCD_10', 'Xrdc_0_EDMA_TCD_11', 'Xrdc_0_Debug_APB_Page0', 'Xrdc_0_Debug_APB_Page1', 'Xrdc_0_Debug_APB_Page2', 'Xrdc_0_Debug_APB_Page3', 'Xrdc_0_Debug_APB_Paged_Area', 'Xrdc_0_SDA_AP', 'Xrdc_0_EIM_0', 'Xrdc_0_ERM_0', 'Xrdc_0_MSCM', 'Xrdc_0_PRAM_0', 'Xrdc_0_PFC', 'Xrdc_0_PFC_alt', 'Xrdc_0_SWT_0', 'Xrdc_0_STM_0', 'Xrdc_0_XRDC', 'Xrdc_0_INTM', 'Xrdc_0_DMAMUX_0', 'Xrdc_0_DMAMUX_1', 'Xrdc_0_RTC', 'Xrdc_0_MC_RGM', 'Xrdc_0_SIUL2_VIRTWRAPPER_PDAC0_HSE_B_0', 'Xrdc_0_SIUL2_VIRTWRAPPER_PDAC0_HSE_B_1', 'Xrdc_0_SIUL2_VIRTWRAPPER_PDAC1_M7_0_0', 'Xrdc_0_SIUL2_VIRTWRAPPER_PDAC1_M7_0_1', 'Xrdc_0_SIUL2_VIRTWRAPPER_PDAC3', 'Xrdc_0_DCM', 'Xrdc_0_WKPU', 'Xrdc_0_CMU_0_5', 'Xrdc_0_TSPC', 'Xrdc_0_SIRC', 'Xrdc_0_SXOSC', 'Xrdc_0_FIRC', 'Xrdc_0_FXOSC', 'Xrdc_0_MC_CGM', 'Xrdc_0_MC_ME', 'Xrdc_0_PLL', 'Xrdc_0_PMC', 'Xrdc_0_FMU', 'Xrdc_0_FMU_alt', 'Xrdc_0_PIT_2', 'Xrdc_0_FlexCAN_0', 'Xrdc_0_FlexCAN_1', 'Xrdc_0_FlexCAN_2', 'Xrdc_0_FlexCAN_3', 'Xrdc_0_FlexCAN_4', 'Xrdc_0_FlexCAN_5', 'Xrdc_0_FlexIO', 'Xrdc_0_LPUART_0', 'Xrdc_0_LPUART_1', 'Xrdc_0_LPUART_2', 'Xrdc_0_LPUART_3', 'Xrdc_0_LPUART_4', 'Xrdc_0_LPUART_5', 'Xrdc_0_LPUART_6', 'Xrdc_0_LPUART_7', 'Xrdc_0_LPI2C_0', 'Xrdc_0_LPI2C_1', 'Xrdc_0_LPSPI_0', 'Xrdc_0_LPSPI_1', 'Xrdc_0_LPSPI_2', 'Xrdc_0_LPSPI_3', 'Xrdc_0_SAI_0', 'Xrdc_0_LPCMP_0', 'Xrdc_0_LPCMP_1', 'Xrdc_0_TMU', 'Xrdc_0_CRC', 'Xrdc_0_FCCU', 'Xrdc_0_MTR', 'Xrdc_0_MU_0', 'Xrdc_0_JDC', 'Xrdc_0_CONFIGURATION_GPR', 'Xrdc_0_STCU', 'Xrdc_0_SELFTEST_GPR', 'Xrdc_0_TCM_XBIC', 'Xrdc_0_EDMA_XBIC', 'Xrdc_0_EDMA_TCD_12', 'Xrdc_0_EDMA_TCD_13', 'Xrdc_0_EDMA_TCD_14', 'Xrdc_0_EDMA_TCD_15', 'Xrdc_0_EDMA_TCD_16', 'Xrdc_0_EDMA_TCD_17', 'Xrdc_0_EDMA_TCD_18', 'Xrdc_0_EDMA_TCD_19', 'Xrdc_0_EDMA_TCD_20', 'Xrdc_0_EDMA_TCD_21', 'Xrdc_0_EDMA_TCD_22', 'Xrdc_0_EDMA_TCD_23', 'Xrdc_0_EDMA_TCD_24', 'Xrdc_0_EDMA_TCD_25', 'Xrdc_0_EDMA_TCD_26', 'Xrdc_0_EDMA_TCD_27', 'Xrdc_0_EDMA_TCD_28', 'Xrdc_0_EDMA_TCD_29', 'Xrdc_0_EDMA_TCD_30', 'Xrdc_0_EDMA_TCD_31', 'Xrdc_0_SEMA_42', 'Xrdc_0_PRAM_1', 'Xrdc_0_STM_1', 'Xrdc_0_EMAC', 'Xrdc_0_LPUART_8', 'Xrdc_0_LPUART_9', 'Xrdc_0_LPUART_10', 'Xrdc_0_LPUART_11', 'Xrdc_0_LPUART_12', 'Xrdc_0_LPUART_13', 'Xrdc_0_LPUART_14', 'Xrdc_0_LPUART_15', 'Xrdc_0_LPSPI_4', 'Xrdc_0_LPSPI_5', 'Xrdc_0_QuadSPI', 'Xrdc_0_SAI_1', 'Xrdc_0_LPCMP_2', 'Xrdc_0_MU_1']

4.45 Parameter XrdcSema4Enable

Enable Semaphore would require the domain must own the gate to have access right for this peripheral

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.46 Reference XrdcSema42LogicChannel

Select XrdcSema42LogicChannel to select specified Sema42 Logic Channel

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/TS_T40D34M20I0R0/Rm/RmConfigSet/Sema42_ModuleConfig/Sema42↩ LogicChannelConfiguration

4.47 Container MPU_M7_Configuration

Configuration for the ARM Cortex M7 MPU.

Included subcontainers:

- MPU_M7_ModuleConfig

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.48 Parameter MpuDevErrorDetect

MpuDevErrorDetect

Switches the Development Error Detection and Notification ON or OFF.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.49 Parameter ClearCacheEnable

Vendor specific: Enable clearing of memory regions that use cache

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.50 Container MPU_M7_ModuleConfig

Configuration for the ARM Cortex M7 MPU.

Included subcontainers:

- RegionConfig

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.51 Parameter DefaultMapEnable

Vendor specific: Switch to indicate if the default memmory map is enabled

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.52 Parameter RunInHFNMIEnable

Vendor specific: Switch to indicate if the MPU is enabled in Hard Fault or NMI

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.53 Parameter MemManageInterruptEnable

Vendor specific: Switch to indicate if the Memory Management Interrupt is enabled

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.54 Parameter PhysicalCoreID

Physical Core ID

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	Cortex_M7_Core0
literals	['Cortex_M7_Core0']

4.55 Container RegionConfig

Vendor specific:

Configuration for interrupt requests.

Warning: This is a precompile configuration.If you uncheck a ISR, you will not be able to enable the respective channel or error functionality at post build time.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	16
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.56 Parameter RegionNumber

Vendor specific:

Region Number.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0
max	15
min	0

4.57 Parameter StartAddress

Vendor specific:

Start Address.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0
max	4294967294
min	0

4.58 Parameter EndAddress

Vendor specific:

End Address.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	4294967295
max	4294967295
min	32

4.59 Parameter RegionSize

Vendor specific:

RegionSize.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	1
max	9223372036854775807
min	-9223372036854775808

4.60 Parameter MemoryType

Vendor specific:

Memory type.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	MPU_M7_MEM_STRONG_ORDER
literals	['MPU_M7_MEM_STRONG_ORDER', 'MPU_M7_MEM_DEVICE_SHA↩ RED', 'MPU_M7_MEM_NORMAL_IO_WR_THROUGH', 'MPU_M7_M↩ EM_NORMAL_IO_WR_BACK1', 'MPU_M7_MEM_NORMAL_IO_NO↩ _CACHE', 'MPU_M7_MEM_NORMAL_IO_WR_BACK2', 'MPU_M7_M↩ EM_DEVICE_NOSHARE', 'MPU_M7_MEM_NORMAL_CACHEABLE']

4.61 Parameter AccessRights

Vendor specific:

Access Rights.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	MPU_M7_PRIV_UNPRIV_NONE
literals	['MPU_M7_PRIV_UNPRIV_NONE', 'MPU_M7_PRIV_RW_UNPRIV_N↩ ONE', 'MPU_M7_PRIV_RWX_UNPRIV_NONE', 'MPU_M7_PRIV_RW↩ _UNPRIV_R', 'MPU_M7_PRIV_RWX_UNPRIV_RX', 'MPU_M7_PRIV↩ _RW_UNPRIV_RW', 'MPU_M7_PRIV_RWX_UNPRIV_RWX', 'MPU_↩ M7_PRIV_R_UNPRIV_NONE', 'MPU_M7_PRIV_RX_UNPRIV_NONE', 'MPU_M7_PRIV_R_UNPRIV_R', 'MPU_M7_PRIV_RX_UNPRIV_RX']

4.62 Parameter OuterCachePolicy

Vendor specific:

Outer Cache Policy.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	MPU_M7_CACHE_POLICY_NO_CACHE
literals	['MPU_M7_CACHE_POLICY_NO_CACHE', 'MPU_M7_CACHE_POLI↩ CY_W_BACK_WR_ALLOCATE', 'MPU_M7_CACHE_POLICY_W_TH↩ ROUGH_NO_W_ALLOCATE', 'MPU_M7_CACHE_POLICY_W_BACK↩ _NO_W_ALLOCATE']

4.63 Parameter InnerCachePolicy

Vendor specific:

Inner Cache Policy.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	MPU_M7_CACHE_POLICY_NO_CACHE
literals	['MPU_M7_CACHE_POLICY_NO_CACHE', 'MPU_M7_CACHE_POLI↩ CY_W_BACK_WR_ALLOCATE', 'MPU_M7_CACHE_POLICY_W_TH↩ ROUGH_NO_W_ALLOCATE', 'MPU_M7_CACHE_POLICY_W_BACK↩ _NO_W_ALLOCATE']

4.64 Parameter SubregionMask

Vendor specific:

Sub Region.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0
max	255
min	0

4.65 Parameter Shareable

Vendor specific: Switch to indicate if the region is shareable

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.66 Container Sema42_ModuleConfig

Configuration for the Semaphores2.

Included subcontainers:

- Sema42LogicChannelConfiguration

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.67 Parameter Sema42DevErrorDetect

Sema42DevErrorDetect

Switches the Development Error Detection and Notification ON or OFF.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.68 Container Sema42LogicChannelConfiguration

Configuration for Sema42 Instance

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD

4.69 Parameter Sema42LogicChannel_LogicName

Vendor specific:

Logic Channel Name.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	SEMA42_LOGIC_CH_0

4.70 Parameter Sema42HardwareInstance

Vendor specific: Select the physical Sema42 Instance.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	SEMA42_INSTANCE0
literals	['SEMA42_INSTANCE0']

4.71 Parameter Sema42HardwareChannel

Vendor specific: Select the physical Sema42 Channel.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	15
min	0

4.72 Container Pflash_Configuration

Configuration for the "Pflash.

Included subcontainers:

- PflashMasterProtection

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.73 Parameter PflashDevErrorDetect

PflashDevErrorDetect

Switches the Development Error Detection and Notification ON or OFF.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.74 Container PflashMasterProtection

Pflash Master Protection

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.75 Parameter PflashControler

Select controller: PLFASH or PLFASH ALTERNATE

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	PFLASH_CTRL
literals	['PFLASH_CTRL', 'PFLASH_ATL_CTRL']

4.76 Parameter PflashMAP

Select master for mode protection

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	PFLASH_M0AP
literals	['PFLASH_M15AP', 'PFLASH_M14AP', 'PFLASH_M13AP', 'PFLASH_↩ M12AP', 'PFLASH_M11AP', 'PFLASH_M10AP', 'PFLASH_M9AP', 'PFLA↩ SH_M8AP', 'PFLASH_M7AP', 'PFLASH_M6AP', 'PFLASH_M5AP', 'PFL↩ ASH_M4AP', 'PFLASH_M2AP', 'PFLASH_M1AP', 'PFLASH_M0AP']

4.77 Parameter PflashMasterAccess

Select access right for each master

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	PFLASH_READ_ALLOWED
literals	['PFLASH_READ_NOT_ALLOWED', 'PFLASH_READ_ALLOWED']

4.78 Container Crossbar_Configuration

Configuration for the Crossbar.

Included subcontainers:

- RmCrossbarInstance

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.79 Parameter CrossbarDevErrorDetect

CrossbarDevErrorDetect

Switches the Development Error Detection and Notification ON or OFF.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.80 Container RmCrossbarInstance

Vendor specific: Configuration of a crossbar instance.

Included subcontainers:

- RmCrossbarHwSlavePort

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	2
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.81 Parameter RmCrossbarHwInstance

Vendor specific:

Select the crossbar switch instance.

Note: For S32K3XX, only XRDC_1 (XRDC Lite Peripheral) is programmable. Also for S32K312, AXBS is not programmable.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	AXBS_1
literals	['AXBS_1']

4.82 Container RmCrossbarHwSlavePort

Hardware slave port configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	7
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.83 Parameter RmSlavePortNumber

Vendor specific:

Slave port number in hardware.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	7
min	0

4.84 Parameter RmCrossbarPrioMaster0

Vendor specific:

Priority of the master 0 on the cross bar.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	7
min	0

4.85 Parameter RmCrossbarPrioMaster1

Vendor specific:

Priority of the master1 on the cross bar.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	7
min	0

4.86 Parameter RmCrossbarPrioMaster2

Vendor specific:

Priority of the master2 on the cross bar.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	2
max	7
min	0

4.87 Parameter RmCrossbarPrioMaster3

Vendor specific:

Priority of the master3 on the cross bar.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	3
max	7
min	0

4.88 Parameter RmCrossbarPrioMaster4

Vendor specific:

Priority of the master4 on the cross bar.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	4
max	7
min	0

4.89 Parameter RmCrossbarPrioMaster5

Vendor specific:

Priority of the master5 on the cross bar.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	7
min	0

4.90 Parameter RmCrossbarPrioMaster6

Vendor specific:

Priority of the master6 on the cross bar.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	7
min	0

4.91 Parameter RmCrossbarPrioMaster7

Vendor specific:

Priority of the master7 on the cross bar.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	7
min	0

4.92 Parameter RmCrossbarEnableLock

Vendor specific: Locks the configuration registers for the respective slave port.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.93 Parameter RmCrossbarHaltLowPrio

Vendor specific: Sets the initial arbitration priority for low power mode requests. Setting this bit will not affect the request

for low power mode from attaining highest priority once it has control of the slave ports. Disabled = The low power mode request has the highest priority for arbitration on this slave port;

Enabled = The low power mode request has the lowest initial priority for arbitration on this slave port

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.94 Parameter RmCrossbarEnablePrioElevM0

Vendor specific: On this slave port, enable priority elevation for master 0. If enabled, the master is able to elevate its priority

to the highest.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.95 Parameter RmCrossbarEnablePrioElevM1

Vendor specific: On this slave port, enable priority elevation for master 1. If enabled, the master is able to elevate its priority

to the highest.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.96 Parameter RmCrossbarEnablePrioElevM2

Vendor specific: On this slave port, enable priority elevation for master 2. If enabled, the master is able to elevate its priority

to the highest.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.97 Parameter RmCrossbarEnablePrioElevM3

Vendor specific: On this slave port, enable priority elevation for master 3. If enabled, the master is able to elevate its priority

to the highest.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.98 Parameter RmCrossbarEnablePrioElevM4

Vendor specific: On this slave port, enable priority elevation for master 4. If enabled, the master is able to elevate its priority

to the highest.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.99 Parameter RmCrossbarEnablePrioElevM5

Vendor specific: On this slave port, enable priority elevation for master 5. If enabled, the master is able to elevate its priority

to the highest.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.100 Parameter RmCrossbarEnablePrioElevM6

Vendor specific: On this slave port, enable priority elevation for master 6. If enabled, the master is able to elevate its priority

to the highest.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.101 Parameter RmCrossbarEnablePrioElevM7

Vendor specific: On this slave port, enable priority elevation for master 7. If enabled, the master is able to elevate its priority

to the highest.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.102 Parameter RmCrossbarEnableFixedPrio

Vendor specific: On this slave port, select the arbitraion mode: if enabled the arbitration mode will be FIXED PRIORITY, if disabled it will be ROUND ROBIN.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.103 Parameter RmCrossbarParkingControl

Determines the slave port?s parking control. The low-power park feature results in an overall power savings if the slave port is not saturated; however, this forces an extra latency clock when any master tries to access the slave port while not in use because it is not parked on any master.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	ParkField
literals	['ParkField', 'LastMaster', 'LowPowerPark']

4.104 Parameter RmCrossbarParkField

Vendor specific:

Determines which master port the current slave port parks on when no masters are actively making requests and the RmCrossbarParkingControl=ParkField.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Master0
literals	['Master0', 'Master1', 'Master2', 'Master3', 'Master4', 'Master5', 'Master6', 'Master7']

4.105 Container Xbic_Configuration

Configuration for the Xbic.

Included subcontainers:

- XbicCheckingControl

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.106 Parameter XbicDevErrorDetect

XbicDevErrorDetect

Switches the Development Error Detection and Notification ON or OFF.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.107 Container XbicCheckingControl

Xbic Checking Control

Configuration enable/disable slave port EDC Error Detection (SE) and master port for Feedback Integrity Check (ME) of all of Xbic Instances.

Note: Any SE and ME that are not configured will be enabled by default

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	4
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.108 Parameter XbicInstance

Select Xbic instance

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	XBIC_0
literals	['XBIC_0', 'XBIC_1', 'XBIC_2', 'XBIC_3']

4.109 Parameter SE0

Enable/Disable Attribute integrity checking for slave port

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE

Property	Value
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.110 Parameter SE1

Enable/Disable Attribute integrity checking for slave port

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.111 Parameter SE2

Enable/Disable Attribute integrity checking for slave port

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.112 Parameter SE3

Enable/Disable Attribute integrity checking for slave port

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.113 Parameter SE4

Enable/Disable Attribute integrity checking for slave port

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.114 Parameter SE5

Enable/Disable Attribute integrity checking for slave port

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.115 Parameter SE6

Enable/Disable Attribute integrity checking for slave port

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.116 Parameter SE7

Enable/Disable Attribute integrity checking for slave port

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.117 Parameter ME0

Enable/Disable Feedback integrity checking for master port

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.118 Parameter ME1

Enable/Disable Feedback integrity checking for master port

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.119 Parameter ME2

Enable/Disable Feedback integrity checking for master port

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.120 Parameter ME3

Enable/Disable Feedback integrity checking for master port

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.121 Parameter ME4

Enable/Disable Feedback integrity checking for master port

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.122 Parameter ME5

Enable/Disable Feedback integrity checking for master port

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.123 Parameter ME6

Enable/Disable Feedback integrity checking for master port

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.124 Parameter ME7

Enable/Disable Feedback integrity checking for master port

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.125 Container Virt_Wrapper_Configuration

Configuration for the "Virtual Wrapper.

Included subcontainers:

- Mscr_Config_List
- Imcr_Config_List
- Other_Config_List

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.126 Parameter VirtWrapperDevErrorDetect

VirtWrapperDevErrorDetect

Switches the Development Error Detection and Notification ON or OFF.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.127 Container Mscr_Config_List

Vendor specific:

Virtual wrraper for MSCR.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	218
upperMultiplicity	218
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.128 Parameter MscrName

Select output pad

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	PTA0
literals	['PTA0', 'PTA1', 'PTA2', 'PTA3', 'PTA4', 'PTA5', 'PTA6', 'PTA7', 'PTA8', 'PTA9', 'PTA10', 'PTA11', 'PTA12', 'PTA13', 'PTA14', 'PTA15', 'PTA16', 'PTA17', 'PTA18', 'PTA19', 'PTA20', 'PTA21', 'PTA22', 'PTA23', 'PTA24', 'PTA25', 'PTA26', 'PTA27', 'PTA28', 'PTA29', 'PTA30', 'PTA31', 'PTB0', 'PTB1', 'PTB2', 'PTB3', 'PTB4', 'PTB5', 'PTB8', 'PTB9', 'PTB10', 'PTB11', 'PTB12', 'PTB13', 'PTB14', 'PTB15', 'PTB16', 'PTB17', 'PTB18', 'PTB19', 'PTB20', 'PTB21', 'PTB22', 'PTB23', 'PTB24', 'PTB25', 'PTB26', 'PTB27', 'PTB28', 'PTB29', 'PTB30', 'PTB31', 'PTC0', 'PTC1', 'PTC2', 'PTC3', 'PTC4', 'PTC5', 'PTC6', 'PTC7', 'PTC8', 'PTC9', 'PTC10', 'PTC11', 'PTC12', 'PTC13', 'PTC14', 'PTC15', 'PTC16', 'PTC17', 'PTC18', 'PTC19', 'PTC20', 'PTC21', 'PTC22', 'PTC23', 'PTC24', 'PTC25', 'PTC26', 'PTC27', 'PTC28', 'PTC29', 'PTC30', 'PTC31', 'PTD0', 'PTD1', 'PTD2', 'PTD3', 'PTD4', 'PTD5', 'PTD6', 'PTD7', 'PTD8', 'PTD9', 'PTD10', 'PTD11', 'PTD12', 'PTD13', 'PTD14', 'PTD15', 'PTD16', 'PTD17', 'PTD18', 'PTD19', 'PTD20', 'PTD21', 'PTD22', 'PTD23', 'PTD24', 'PTD25', 'PTD26', 'PTD27', 'PTD28', 'PTD29', 'PTD30', 'PTD31', 'PTE0', 'PTE1', 'PTE2', 'PTE3', 'PTE4', 'PTE5', 'PTE6', 'PTE7', 'PTE8', 'PTE9', 'PTE10', 'PTE11', 'PTE12', 'PTE13', 'PTE14', 'PTE15', 'PTE16', 'PTE17', 'PTE18', 'PTE19', 'PTE20', 'PTE21', 'PTE22', 'PTE23', 'PTE24', 'PTE25', 'PTE26', 'PTE27', 'PTE28', 'PTE29', 'PTE30', 'PTE31', 'PTF0', 'PTF1', 'PTF2', 'PTF3', 'PTF4', 'PTF5', 'PTF6', 'PTF7', 'PTF8', 'PTF9', 'PTF10', 'PTF11', 'PTF12', 'PTF13', 'PTF14', 'PTF15', 'PTF16', 'PTF17', 'PTF18', 'PTF19', 'PTF20', 'PTF21', 'PTF22', 'PTF23', 'PTF24', 'PTF25', 'PTF26', 'PTF27', 'PTF28', 'PTF29', 'PTF30', 'PTF31', 'PTG0', 'PTG1', 'PTG2', 'PTG3', 'PTG4', 'PTG5', 'PTG6', 'PTG7', 'PTG8', 'PTG9', 'PTG10', 'PTG11', 'PTG12', 'PTG13', 'PTG14', 'PTG15', 'PTG16', 'PTG17', 'PTG18', 'PTG19', 'PTG20', 'PTG21', 'PTG22', 'PTG23', 'PTG24', 'PTG25', 'PTG26', 'PTG27']

4.129 Parameter MscrNumber

Seclect pad number

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	False

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	65535
min	0

4.130 Parameter MscrSiul2Instance

Select SIUL2 Instance

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	SIUL2_0
literals	['SIUL2_0']

4.131 Parameter MscrPinMux

Select Pin Mux

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	GPIO_0

4.132 Parameter MscrMirror

Select SIUL2 Mirror

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Xrdc_0_SIUL2_VIRTWRAPPER_PDAC0_HSE_B_0
literals	['Xrdc_0_SIUL2_VIRTWRAPPER_PDAC0_HSE_B_0', 'Xrdc_0_SIUL2_↩ VIRTWRAPPER_PDAC0_HSE_B_1', 'Xrdc_0_SIUL2_VIRTWRAPPER↩ _PDAC1_M7_0_0', 'Xrdc_0_SIUL2_VIRTWRAPPER_PDAC1_M7_0_1']

4.133 Container Imcr_Config_List

Vendor specific:

IMCR configuration list.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	288
upperMultiplicity	288
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.134 Parameter ImcrName

Select input register

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	SIUL2_0_IMCR0

Property	Value
literals	['SIUL2_0_IMCR0', 'SIUL2_0_IMCR1', 'SIUL2_0_IMCR2', 'SIUL2_0_IMCR3', 'SIUL2_0_IMCR4', 'SIUL2_0_IMCR5', 'SIUL2_0_IMCR16', 'SIUL2_0_IMCR17', 'SIUL2_0_IMCR18', 'SIUL2_0_IMCR19', 'SIUL2_0_IMCR20', 'SIUL2_0_IMCR21', 'SIUL2_0_IMCR22', 'SIUL2_0_IMCR23', 'SIUL2_0_IMCR24', 'SIUL2_0_IMCR25', 'SIUL2_0_IMCR26', 'SIUL2_0_IMCR27', 'SIUL2_0_IMCR28', 'SIUL2_0_IMCR29', 'SIUL2_0_IMCR30', 'SIUL2_0_IMCR31', 'SIUL2_0_IMCR32', 'SIUL2_0_IMCR33', 'SIUL2_0_IMCR34', 'SIUL2_0_IMCR35', 'SIUL2_0_IMCR36', 'SIUL2_0_IMCR37', 'SIUL2_0_IMCR38', 'SIUL2_0_IMCR39', 'SIUL2_0_IMCR40', 'SIUL2_0_IMCR41', 'SIUL2_0_IMCR42', 'SIUL2_0_IMCR43', 'SIUL2_0_IMCR44', 'SIUL2_0_IMCR45', 'SIUL2_0_IMCR46', 'SIUL2_0_IMCR47', 'SIUL2_0_IMCR48', 'SIUL2_0_IMCR49', 'SIUL2_0_IMCR50', 'SIUL2_0_IMCR51', 'SIUL2_0_IMCR52', 'SIUL2_0_IMCR53', 'SIUL2_0_IMCR54', 'SIUL2_0_IMCR55', 'SIUL2_0_IMCR56', 'SIUL2_0_IMCR57', 'SIUL2_0_IMCR58', 'SIUL2_0_IMCR59', 'SIUL2_0_IMCR60', 'SIUL2_0_IMCR61', 'SIUL2_0_IMCR62', 'SIUL2_0_IMCR63', 'SIUL2_0_IMCR64', 'SIUL2_0_IMCR65', 'SIUL2_0_IMCR66', 'SIUL2_0_IMCR67', 'SIUL2_0_IMCR68', 'SIUL2_0_IMCR69', 'SIUL2_0_IMCR70', 'SIUL2_0_IMCR71', 'SIUL2_0_IMCR80', 'SIUL2_0_IMCR81', 'SIUL2_0_IMCR82', 'SIUL2_0_IMCR83', 'SIUL2_0_IMCR84', 'SIUL2_0_IMCR85', 'SIUL2_0_IMCR86', 'SIUL2_0_IMCR87', 'SIUL2_0_IMCR88', 'SIUL2_0_IMCR89', 'SIUL2_0_IMCR90', 'SIUL2_0_IMCR91', 'SIUL2_0_IMCR92', 'SIUL2_0_IMCR93', 'SIUL2_0_IMCR94', 'SIUL2_0_IMCR95', 'SIUL2_0_IMCR96', 'SIUL2_0_IMCR97', 'SIUL2_0_IMCR98', 'SIUL2_0_IMCR99', 'SIUL2_0_IMCR100', 'SIUL2_0_IMCR101', 'SIUL2_0_IMCR102', 'SIUL2_0_IMCR103', 'SIUL2_0_IMCR112', 'SIUL2_0_IMCR113', 'SIUL2_0_IMCR114', 'SIUL2_0_IMCR115', 'SIUL2_0_IMCR116', 'SIUL2_0_IMCR117', 'SIUL2_0_IMCR118', 'SIUL2_0_IMCR119', 'SIUL2_0_IMCR120', 'SIUL2_0_IMCR121', 'SIUL2_0_IMCR122', 'SIUL2_0_IMCR123', 'SIUL2_0_IMCR124', 'SIUL2_0_IMCR125', 'SIUL2_0_IMCR126', 'SIUL2_0_IMCR127', 'SIUL2_0_IMCR128', 'SIUL2_0_IMCR129', 'SIUL2_0_IMCR130', 'SIUL2_0_IMCR131', 'SIUL2_0_IMCR132', 'SIUL2_0_IMCR133', 'SIUL2_0_IMCR134', 'SIUL2_0_IMCR135', 'SIUL2_0_IMCR144', 'SIUL2_0_IMCR145', 'SIUL2_0_IMCR146', 'SIUL2_0_IMCR147', 'SIUL2_0_IMCR148', 'SIUL2_0_IMCR149', 'SIUL2_0_IMCR152', 'SIUL2_0_IMCR153', 'SIUL2_0_IMCR154', 'SIUL2_0_IMCR155', 'SIUL2_0_IMCR156', 'SIUL2_0_IMCR157', 'SIUL2_0_IMCR158', 'SIUL2_0_IMCR159', 'SIUL2_0_IMCR160', 'SIUL2_0_IMCR161', 'SIUL2_0_IMCR162', 'SIUL2_0_IMCR163', 'SIUL2_0_IMCR164', 'SIUL2_0_IMCR165', 'SIUL2_0_IMCR166', 'SIUL2_0_IMCR167', 'SIUL2_0_IMCR168', 'SIUL2_0_IMCR169', 'SIUL2_0_IMCR170', 'SIUL2_0_IMCR171', 'SIUL2_0_IMCR172', 'SIUL2_0_IMCR173', 'SIUL2_0_IMCR174', 'SIUL2_0_IMCR175', 'SIUL2_0_IMCR176', 'SIUL2_0_IMCR177', 'SIUL2_0_IMCR178', 'SIUL2_0_IMCR179', 'SIUL2_0_IMCR180', 'SIUL2_0_IMCR181', 'SIUL2_0_IMCR182', 'SIUL2_0_IMCR183', 'SIUL2_0_IMCR184', 'SIUL2_0_IMCR185', 'SIUL2_0_IMCR186', 'SIUL2_0_IMCR187', 'SIUL2_0_IMCR188', 'SIUL2_0_IMCR189', 'SIUL2_0_IMCR190', 'SIUL2_0_IMCR191', 'SIUL2_0_IMCR192', 'SIUL2_0_IMCR193', 'SIUL2_0_IMCR194', 'SIUL2_0_IMCR195', 'SIUL2_0_IMCR196', 'SIUL2_0_IMCR197', 'SIUL2_0_IMCR198', 'SIUL2_0_IMCR199', 'SIUL2_0_IMCR200', 'SIUL2_0_IMCR201', 'SIUL2_0_IMCR202', 'SIUL2_0_IMCR211', 'SIUL2_0_IMCR212', 'SIUL2_0_IMCR213', 'SIUL2_0_IMCR214', 'SIUL2_0_IMCR215', 'SIUL2_0_IMCR216', 'SIUL2_0_IMCR217', 'SIUL2_0_IMCR218', 'SIUL2_0_IMCR219', 'SIUL2_0_IMCR220', 'SIUL2_0_IMCR221', 'SIUL2_0_IMCR222', 'SIUL2_0_IMCR223', 'SIUL2_0_IMCR224', 'SIUL2_0_IMCR225', 'SIUL2_0_IMCR226', 'SIUL2_0_IMCR227', 'SIUL2_0_IMCR228', 'SIUL2_0_IMCR229', 'SIUL2_0_IMCR230', 'SIUL2_0_IMCR231', 'SIUL2_0_IMCR232', 'SIUL2_0_IMCR233', 'SIUL2_0_IMCR234', 'SIUL2_0_IMCR235', 'SIUL2_0_IMCR236', 'SIUL2_0_IMCR237', 'SIUL2_0_IMCR238', 'SIUL2_0_IMCR239', 'SIUL2_0_IMCR240', 'SIUL2_0_IMCR241', 'SIUL2_0_IMC

Property	Value

4.135 Parameter ImcrNumber

Seclect pad number

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	65535
min	0

4.136 Parameter ImcrSiul2Instance

Select SIUL2 Instance

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	SIUL2_0
literals	['SIUL2_0']

4.137 Parameter ImcrInput

Input signal controlled by IMCR

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	CAN0_RX

4.138 Parameter ImcrPad

Input pad controlled by IMCR

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	PTA6
literals	['PTA6', 'PTA28', 'PTB0', 'PTC2', 'PTF21']

4.139 Parameter ImcrMirror

Select SIUL2 Mirror

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Xrdc_0_SIUL2_VIRTWRAPPER_PDAC0_HSE_B_0
literals	['Xrdc_0_SIUL2_VIRTWRAPPER_PDAC0_HSE_B_0', 'Xrdc_0_SIUL2_↩ VIRTWRAPPER_PDAC0_HSE_B_1', 'Xrdc_0_SIUL2_VIRTWRAPPER↩ _PDAC1_M7_0_0', 'Xrdc_0_SIUL2_VIRTWRAPPER_PDAC1_M7_0_1']

4.140 Container Other_Config_List

Vendor specific:

All data need to configure Peripheral for one Domain.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	2
upperMultiplicity	2
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.141 Parameter OtherName

Select input register

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Siul2_0_IntCtrl
literals	['Siul2_0_IntCtrl', 'Siul2_0_RegGCR']

4.142 Parameter OtherNumber

Seclect pad number

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	65535
min	0

4.143 Parameter OthersSiul2Instance

Select SIUL2 Instance

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	SIUL2_0
literals	['SIUL2_0']

4.144 Parameter OtherInput

Registers controlled

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	InterruptRegisters

4.145 Parameter OtherMirror

Select SIUL2 Mirror

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Xrdc_0_SIUL2_VIRTWRAPPER_PDAC0_HSE_B_0
literals	['Xrdc_0_SIUL2_VIRTWRAPPER_PDAC0_HSE_B_0', 'Xrdc_0_SIUL2_↩ VIRTWRAPPER_PDAC0_HSE_B_1', 'Xrdc_0_SIUL2_VIRTWRAPPER↩ _PDAC1_M7_0_0', 'Xrdc_0_SIUL2_VIRTWRAPPER_PDAC1_M7_0_1']

4.146 Container CommonPublishedInformation

Vendor specific:

Common container, aggregated by all modules. It contains published information about vendor and versions.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.147 Parameter ArReleaseMajorVersion

Vendor specific:

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.148 Parameter ArReleaseMinorVersion

Vendor specific:

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.149 Parameter ArReleaseRevisionVersion

Vendor specific:

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.150 Parameter ModuleId

Vendor specific:

Module ID of this module from Module List.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	255
max	255
min	255

4.151 Parameter SwMajorVersion

Vendor specific:

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	2
max	2
min	2

4.152 Parameter SwMinorVersion

Vendor specific:

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.153 Parameter SwPatchVersion

Vendor specific:

Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.154 Parameter VendorApiInfix

Vendor specific:

In driver modules which can be instantiated several times on a single ECU, BSW00347 requires

that the name of APIs is extended by the VendorId and a vendor specific name.

This parameter is used to specify the vendor specific name. In total, the

implementation specific name is generated as follows: <ModuleName>_>VendorId>_<VendorAp

E.g. assuming that the VendorId of the implementor is 123 and the implementer chose a VendorApiInfix of "v11r456" a api name Can_Write defined in the SWS will translate to Can_123_v11r456Write.

This parameter is mandatory for all modules with upper multiplicity > 1. It shall not be used for modules with upper multiplicity =1.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	

4.155 Parameter VendorId

Vendor specific:

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	43
max	43
min	43

Chapter 5

Module Index

5.1 Software Specification

Here is a list of all modules:

AXBS IPV Driver 111
RM Driver 116
Mpu M7 IPV Driver 131
Pflash IPV Driver 142
Sema42 IPV Driver 146
Virt Wrapper IPV Driver 158
XBIC IPV Driver 160
XRDC IPV Driver 169

Chapter 6

Module Documentation

6.1 AXBS IPV Driver

6.1.1 Detailed Description

Data Structures

- struct Axbs_Ip_SlavePortConfigType
  Configuration structure for Slave ports. *More...*
- struct Axbs_Ip_ConfigType
  Configuration structure for Axbs Ip. *More...*

Types Reference

- typedef uint8 Axbs_Ip_PortNumberType
- typedef uint32 Axbs_Ip_InstanceNumberType
- typedef uint32 Axbs_Ip_PortPriorityType
- typedef uint32 Axbs_Ip_PortControlType
- typedef uint8 Axbs_Ip_CrossbarPortType
  Axbs port type.

Function Reference

- void Axbs_Ip_Init (const Axbs_Ip_ConfigType *pAxbsConfig)
  The function initializes AXBS module.
- void Axbs_Ip_DeInit (const Axbs_Ip_ConfigType *pAxbsConfig)
  This function de-initializes the AXBS module.
- void Axbs_Ip_Crs_Write (uint32 u32Addr, uint32 value)
  This function write into Crs register.
- uint32 Axbs_Ip_Crs_Read (uint32 u32Addr)
  This function return the value of Crs register.
- void Axbs_Ip_Prs_Write (uint32 u32Addr, uint32 value)
  This function write into Prs register.

Variables

- const Axbs_Ip_ConfigType Axbs_Config
  Export AXBS configurations.

6.1.2 Data Structure Documentation

6.1.2.1 struct Axbs_Ip_SlavePortConfigType

Configuration structure for Slave ports.

Definition at line 132 of file Axbs_Ip_Types.h.

Data Fields

Type	Name	Description
const Axbs_Ip_PortNumberType	Axbs_PortNumber	hardware port number
const Axbs_Ip_InstanceNumberType	Axbs_InstanceNumber	hardware instance
const Axbs_Ip_PortPriorityType	Axbs_PortPriorityConfig	port priority config
const Axbs_Ip_PortControlType	Axbs_PortControlConfig	port control config

6.1.2.2 struct Axbs_Ip_ConfigType

Configuration structure for Axbs Ip.

Definition at line 148 of file Axbs_Ip_Types.h.

Data Fields

Type	Name	Description
const Axbs_Ip_CrossbarPortType	AxbsCrossbarIpNumPorts	The number of configured channels.
const Axbs_Ip_SlavePortConfigType(*const	pAxbsCrossbarSlaveHwIpConfig)[]	configuration of the crossbar IP for Slave

6.1.3 Types Reference

6.1.3.1 Axbs_Ip_PortNumberType

```
typedef uint8 Axbs_Ip_PortNumberType
```

Axbs port number type

Definition at line 108 of file Axbs_Ip_Types.h.

6.1.3.2 Axbs_Ip_InstanceNumberType

```
typedef uint32 Axbs_Ip_InstanceNumberType
```

Axbs instance number type

Definition at line 112 of file Axbs_Ip_Types.h.

6.1.3.3 Axbs_Ip_PortPriorityType

```
typedef uint32 Axbs_Ip_PortPriorityType
```

Axbs port priority config

Definition at line 116 of file Axbs_Ip_Types.h.

6.1.3.4 Axbs_Ip_PortControlType

```
typedef uint32 Axbs_Ip_PortControlType
```

Axbs port control type

Definition at line 120 of file Axbs_Ip_Types.h.

6.1.3.5 Axbs_Ip_CrossbarPortType

```
typedef uint8 Axbs_Ip_CrossbarPortType
```

Axbs port type.

Definition at line 124 of file Axbs_Ip_Types.h.

6.1.4 Function Reference

6.1.4.1 Axbs_Ip_Init()

```
void Axbs_Ip_Init (
            const Axbs_Ip_ConfigType * pAxbsConfig )
```

The function initializes AXBS module.

The function sets the internal module variables according to given configuration set.

Parameters

in	*pAxbsConfig*	Pointer of Configuration Axbs

Returns

Precondition

6.1.4.2 Axbs_Ip_DeInit()

```
void Axbs_Ip_DeInit (
            const Axbs_Ip_ConfigType * pAxbsConfig )
```

This function de-initializes the AXBS module.

Parameters

in	*pAxbsConfig*	Pointer to Axbs configuration structure

6.1.4.3 Axbs_Ip_Crs_Write()

```
void Axbs_Ip_Crs_Write (
            uint32 u32Addr,
            uint32 value )
```

This function write into Crs register.

Parameters

in	*u32Addr*	The Crs register.
in	*value*	The value will be written.

6.1.4.4 Axbs_Ip_Crs_Read()

```
uint32 Axbs_Ip_Crs_Read (
            uint32 u32Addr )
```

This function return the value of Crs register.

Parameters

in	*u32Addr*	The Crs register.

Returns

the value of Crs register.

6.1.4.5 Axbs_Ip_Prs_Write()

```
void Axbs_Ip_Prs_Write (
            uint32 u32Addr,
            uint32 value )
```

This function write into Prs register.

6.1.5 Variable Documentation

6.1.5.1 Axbs_Config

```
const Axbs_Ip_ConfigType Axbs_Config  [extern]
```

Export AXBS configurations.

6.2 RM Driver

6.2.1 Detailed Description

Types Reference

- typedef Xrdc_Ip_SecureAttributeType Rm_XrdcSecureAttributeType
- typedef Xrdc_Ip_DomainIDErrStatusType Rm_XrdcDomainIDErrStatusType
- typedef Xrdc_Ip_MasterCoreType Rm_XrdcMasterCoreType
- typedef uint32 Rm_DomainIdType
- typedef uint8 Rm_Sema42_ChannelType
- typedef uint8 Rm_Sema42_CoreType
- typedef Sema42_Ip_StatusType Rm_Sema42_StatusType
- typedef Mpu_M7_Ip_RegionConfigType Rm_Mpu_M7_RegionConfigType
- typedef Mpu_M7_Ip_AccessRightsType Rm_Mpu_M7_AccessRightsType
- typedef Mpu_M7_Ip_ErrorDetailsType Rm_Mpu_M7_ErrorDetailsType
- typedef Xbic_Ip_ErrorInfoType Rm_XbicErrorInfoType

Function Reference

- void Rm_Init (Rm_ConfigType const *ConfigPtr)

  This function initializes the RM hardware components.
- void Rm_XrdcSetProcessID (Rm_XrdcMasterCoreType MasterCore, uint8 u8PID, Rm_XrdcSecureAttributeType SecureAttr)

  Write process ID for specific master core.
- Rm_DomainIdType Rm_XrdcGetDomainID (uint32 u32Instance)

  Retrieves current domain ID.
- void Rm_XrdcGetDomainIDErrorStatus (uint32 u32Instance, Rm_XrdcDomainIDErrStatusType *pXrdc↩ DomainIDErrorStatus)

  Retrieve domain ID error list.
- void Rm_XrdcInstanceInit (Rm_XrdcInstanceConfigType const *pXrdcInstanceConfig)

  Initialize XRDC instance.
- Rm_Sema42_CoreType Rm_SemaphoreGetStatus (Rm_Sema42_ChannelType ChannelNumber)

  Rm_SemaphoreGetStatus.
- Std_ReturnType Rm_SemaphoreLockGate (Rm_Sema42_ChannelType ChannelNumber)

  Rm_SemaphoreLockGate.
- Std_ReturnType Rm_SemaphoreUnlockGate (Rm_Sema42_ChannelType ChannelNumber)

  Rm_SemaphoreUnlockGate.
- void Rm_SemaphoreResetGate (Rm_Sema42_ChannelType ChannelNumber)

  Rm_SemaphoreResetGate.
- void Rm_SemaphoreResetAllGates (void)

  Rm_SemaphoreResetAllGates.
- void Rm_Mpu_M7_SetRegionConfig (uint8 u8RegionNum, const Rm_Mpu_M7_RegionConfigType *const pUserConfigPtr)

  Configures the region selected by u8RegionNum with the data from pUserConfigPtr.
- void Rm_Mpu_M7_EnableRegion (uint8 u8RegionNum, boolean bEnable)

Enables or disabled a specific region.

- void Rm_Mpu_M7_SetAccessRight (uint8 u8RegionNum, Rm_Mpu_M7_AccessRightsType eRights)

  Modify the access rights for a specific region.
- Std_ReturnType Rm_Mpu_M7_GetErrorDetails (Rm_Mpu_M7_ErrorDetailsType *pErrorDetails)

  Retrieve error details.
- void Rm_Mpu_M7_MulticoreInit (void)

  Disables the module and resets all region configurations.
- void Rm_XbicEnableMasterFeedbackCheck (uint32 xbicInstance, uint8 masterPort, boolean bFeedback↩CheckEnable)

  Enable/Disable feedback integrity checking for a specific master port.
- void Rm_XbicEnableSlaveEDCCheck (uint32 xbicInstance, uint8 slavePort, boolean bEDCCheckEnable)

  Enable/Disable attribute integrity checking for a specific slave port.
- void Rm_XbicFeedbackCheckAndEDCCheckDisable (void)

  Disable all of feedback integrity checking for the master ports and attribute integrity checking for slave ports.
- void Rm_XbicGetErrorStatus (uint32 u32XbicInstance, Rm_XbicErrorInfoType *pErrorInfo)

  Get error status of a Xbic instance.
- void Rm_XbicErrorInjection (uint32 u32XbicInstance, uint8 slaveTarget, uint8 masterTargetID, uint8 syn↩Decode)

  Inject an error into a master, a slave of a Xbic instance with a specific error code.
- void Rm_XbicErrorInjectionDisable (uint32 u32XbicInstance)

  Disable Inject error a specific Xbic instance.
- void Rm_GetVersionInfo (Std_VersionInfoType *versioninfo)

  This service returns the version information of this module.

Variables

- const Rm_ConfigType Rm_Config

6.2.2 Types Reference

6.2.2.1 Rm_XrdcSecureAttributeType

```
typedef Xrdc_Ip_SecureAttributeType Rm_XrdcSecureAttributeType
```

Enumeration listing secure attributes of a XRDC master

Definition at line 259 of file CDD_Rm_Ipw.h.

6.2.2.2 Rm_XrdcDomainIDErrStatusType

```
typedef Xrdc_Ip_DomainIDErrStatusType Rm_XrdcDomainIDErrStatusType
```

Structure used to retrieve information violation details and the domain ID where violation occured.

Definition at line 262 of file CDD_Rm_Ipw.h.

6.2.2.3 Rm_XrdcMasterCoreType

```
typedef Xrdc_Ip_MasterCoreType Rm_XrdcMasterCoreType
```

Master core type, use for PID feature

Definition at line 265 of file CDD_Rm_Ipw.h.

6.2.2.4 Rm_DomainIdType

```
typedef uint32 Rm_DomainIdType
```

Domain ID

Definition at line 268 of file CDD_Rm_Ipw.h.

6.2.2.5 Rm_Sema42_ChannelType

```
typedef uint8 Rm_Sema42_ChannelType
```

Channel Type for Sema42

Definition at line 275 of file CDD_Rm_Ipw.h.

6.2.2.6 Rm_Sema42_CoreType

```
typedef uint8 Rm_Sema42_CoreType
```

Core Type for Sema42

Definition at line 278 of file CDD_Rm_Ipw.h.

6.2.2.7 Rm_Sema42_StatusType

```
typedef Sema42_Ip_StatusType Rm_Sema42_StatusType
```

Status Type for Sema42

Definition at line 281 of file CDD_Rm_Ipw.h.

6.2.2.8 Rm_Mpu_M7_RegionConfigType

```
typedef Mpu_M7_Ip_RegionConfigType Rm_Mpu_M7_RegionConfigType
```

Configuration structure containing the region configuration for Mpu M7

Definition at line 286 of file CDD_Rm_Ipw.h.

6.2.2.9 Rm_Mpu_M7_AccessRightsType

```
typedef Mpu_M7_Ip_AccessRightsType Rm_Mpu_M7_AccessRightsType
```

Enumeration listing access permisions for Mpu M7

Definition at line 289 of file CDD_Rm_Ipw.h.

6.2.2.10 Rm_Mpu_M7_ErrorDetailsType

```
typedef Mpu_M7_Ip_ErrorDetailsType Rm_Mpu_M7_ErrorDetailsType
```

Enumeration listing the possible access errors for Mpu M7

Definition at line 292 of file CDD_Rm_Ipw.h.

6.2.2.11 Rm_XbicErrorInfoType

```
typedef Xbic_Ip_ErrorInfoType Rm_XbicErrorInfoType
```

Structure for error informations of Xbic Ip.

Definition at line 297 of file CDD_Rm_Ipw.h.

6.2.3 Function Reference

6.2.3.1 Rm_Init()

```
void Rm_Init (
            Rm_ConfigType const * ConfigPtr )
```

This function initializes the RM hardware components.

This service is a non reentrant function used for driver initialization. The Initialization function shall initialize all relevant registers of the configured hardware with the values of the structure referenced by the parameter Config↩ Ptr. If the hardware allows for only one usage of the register, the driver module implementing that functionality is responsible for initializing the register. The initialization function of this module shall always have a pointer as a parameter, even though for Variant PC no configuration set shall be given. Instead a NULL pointer shall be passed to the initialization function.

Parameters

in	*ConfigPtr*	Pointer to a selected configuration structure.

Returns

void

6.2.3.2 Rm_XrdcSetProcessID()

```
void Rm_XrdcSetProcessID (
            Rm_XrdcMasterCoreType MasterCore,
            uint8 u8PID,
            Rm_XrdcSecureAttributeType SecureAttr )
```

Write process ID for specific master core.

Parameters

in	*MasterCore*	Master core instance
in	*u8PID*	Processor ID
in	*SecureAttr*	XRDC_SECURE XRDC_NON_SECURE
out	*none*	

Returns

none

6.2.3.3 Rm_XrdcGetDomainID()

```
Rm_DomainIdType Rm_XrdcGetDomainID (
            uint32 u32Instance )
```

Retrieves current domain ID.

Parameters

in	*u32Instance*	XRDC instance
out	*none*	

Returns

Rm_DomainIdType Domain ID

6.2.3.4 Rm_XrdcGetDomainIDErrorStatus()

```
void Rm_XrdcGetDomainIDErrorStatus (
            uint32 u32Instance,
            Rm_XrdcDomainIDErrStatusType * pXrdcDomainIDErrorStatus )
```

Retrieve domain ID error list.

Parameters

in	*u32Instance*	XRDC instance
in	*pXrdcDomainIDErrorStatus*	uint32 DomainIDAccessError Xrdc_Ip_ErrorStatusType ErrorStatus[20]
out	*none*	

Returns

none

6.2.3.5 Rm_XrdcInstanceInit()

```
void Rm_XrdcInstanceInit (
            Rm_XrdcInstanceConfigType const * pXrdcInstanceConfig )
```

Initialize XRDC instance.

Parameters

in	*uint32*	pXrdcInstanceConfig - XRDC instance configuration
out	*none*	

Returns

none

6.2.3.6 Rm_SemaphoreGetStatus()

```
Rm_Sema42_CoreType Rm_SemaphoreGetStatus (
            Rm_Sema42_ChannelType ChannelNumber )
```

Rm_SemaphoreGetStatus.

This function returns the status of the selected Semaphore. The return is either the number of the core that keeps it locked, or RM_SEMAPHORE_FREE (255) if the Semaphore is free

Parameters

in	*ChannelNumber*	Semaphore channel

Return values

Returns	the bumber of the core if the semaphore is locked and RM_SEMAPHORE_FREE (255)

6.2.3.7 Rm_SemaphoreLockGate()

```
Std_ReturnType Rm_SemaphoreLockGate (
            Rm_Sema42_ChannelType ChannelNumber )
```

Rm_SemaphoreLockGate.

This function locks a Semaphore from a given core, in order to guard a critical section

Parameters

in	*ChannelNumber*	Semaphore channel

Returns

Std_ReturnType

Return values

E_OK	The function call is valid
E_NOT_OK	The function call is invalid

6.2.3.8 Rm_SemaphoreUnlockGate()

```
Std_ReturnType Rm_SemaphoreUnlockGate (
            Rm_Sema42_ChannelType ChannelNumber )
```

Rm_SemaphoreUnlockGate.

This function unlocks a Semaphore from a given core, in order to mark the end of a critical section

Parameters

in	*ChannelNumber*	Semaphore channel

Returns

Std_ReturnType

Return values

E_OK	The function call is valid
E_NOT_OK	The function call is invalid

6.2.3.9 Rm_SemaphoreResetGate()

```
void Rm_SemaphoreResetGate (
            Rm_Sema42_ChannelType ChannelNumber )
```

Rm_SemaphoreResetGate.

This function Resets the status of all Semaphores configured by the RM driver The API is not part of the standard semaphore usage, it should only be used in case of faults in the hardware or a deadlock.

Parameters

in	*ChannelNumber*	Semaphore channel

6.2.3.10 Rm_SemaphoreResetAllGates()

```
void Rm_SemaphoreResetAllGates (
            void  )
```

Rm_SemaphoreResetAllGates.

This function Resets the status of all Semaphores configured by the RM driver. The API is not part of the standard semaphore usage, it should only be used in case of faults in the hardware or a deadlock.

Parameters

in	*NA*	

6.2.3.11 Rm_Mpu_M7_SetRegionConfig()

```
void Rm_Mpu_M7_SetRegionConfig (
            uint8 u8RegionNum,
            const Rm_Mpu_M7_RegionConfigType *const pUserConfigPtr )
```

Configures the region selected by u8RegionNum with the data from pUserConfigPtr.

This function is Reentrant

Parameters

in	*u8RegionNum*	region number
in	*pUserConfigPtr*	pointer to the region configuration structure for MPU M7.

Returns

void

Precondition

6.2.3.12 Rm_Mpu_M7_EnableRegion()

```
void Rm_Mpu_M7_EnableRegion (
            uint8 u8RegionNum,
            boolean bEnable )
```

Enables or disabled a specific region.

This function is Reentrant

Parameters

in	*u8RegionNum*	Region to be modified
in	*bEnable*	Specifies whether the region is enabled or disabled

Returns

void

Precondition

6.2.3.13 Rm_Mpu_M7_SetAccessRight()

```
void Rm_Mpu_M7_SetAccessRight (
            uint8 u8RegionNum,
            Rm_Mpu_M7_AccessRightsType eRights )
```

Modify the access rights for a specific region.

This function is Reentrant

Parameters

in	*u8RegionNum*	Region to be modified
in	*eRights*	Specifies the new access rights

Returns

void

Precondition

6.2.3.14 Rm_Mpu_M7_GetErrorDetails()

```
Std_ReturnType Rm_Mpu_M7_GetErrorDetails (
            Rm_Mpu_M7_ErrorDetailsType * pErrorDetails )
```

Retrieve error details.

This function is Reentrant

Parameters

out	*pErrorDetails*	: Storage where the data will be saved

Returns

Std_ReturnType

Return values

E_OK	if an error was present
E_NOT_OK	otherwise

Precondition

6.2.3.15 Rm_Mpu_M7_MulticoreInit()

```
void Rm_Mpu_M7_MulticoreInit (
            void  )
```

Disables the module and resets all region configurations.

This function is Reentrant

Returns

void

Precondition

6.2.3.16 Rm_XbicEnableMasterFeedbackCheck()

```
void Rm_XbicEnableMasterFeedbackCheck (
            uint32 xbicInstance,
            uint8 masterPort,
            boolean bFeedbackCheckEnable )
```

Enable/Disable feedback integrity checking for a specific master port.

This function is Reentrant.

Parameters

in	*xbicInstance*	: Xbic instance number
in	*masterPort*	: Master port number
in	*bFeedbackCheckEnable*	: Specifies whether the feedback integrity checking for the master port is enabled or disabled

Returns

void

Precondition

6.2.3.17 Rm_XbicEnableSlaveEDCCheck()

```
void Rm_XbicEnableSlaveEDCCheck (
            uint32 xbicInstance,
            uint8 slavePort,
            boolean bEDCCheckEnable )
```

Enable/Disable attribute integrity checking for a specific slave port.

This function is Reentrant.

Parameters

in	*xbicInstance*	: Xbic instance number
in	*slavePort*	: Slave port number
in	*bEDCCheckEnable*	: Specifies whether the attribute integrity checking for the slave port is enabled or disabled

Returns

void

Precondition

6.2.3.18 Rm_XbicFeedbackCheckAndEDCCheckDisable()

```
void Rm_XbicFeedbackCheckAndEDCCheckDisable (
            void  )
```

Disable all of feedback integrity checking for the master ports and attribute integrity checking for slave ports.

This function is Reentrant.

Returns

void

Precondition

6.2.3.19 Rm_XbicGetErrorStatus()

```
void Rm_XbicGetErrorStatus (
            uint32 u32XbicInstance,
            Rm_XbicErrorInfoType * pErrorInfo )
```

Get error status of a Xbic instance.

This function is Reentrant.

Parameters

in	*xbicInstance*	: Xbic instance number
in	*pErrorInfo*	: Pointer points to the memory area containing the XBIC reports error information

Returns

void

Precondition

6.2.3.20 Rm_XbicErrorInjection()

```
void Rm_XbicErrorInjection (
            uint32 u32XbicInstance,
            uint8 slaveTarget,
            uint8 masterTargetID,
            uint8 synDecode )
```

Inject an error into a master, a slave of a Xbic instance with a specific error code.

This function is Reentrant.

Parameters

in	*xbicInstance*	: Xbic instance number
in	*slavePort*	: Slave port number injected error
in	*masterTargetID*	: Master tagget ID number injected error
in	*synDecode*	: Error Syndrome injected

Returns

void

Precondition

6.2.3.21 Rm_XbicErrorInjectionDisable()

```
void Rm_XbicErrorInjectionDisable (
            uint32 u32XbicInstance )
```

Disable Inject error a specific Xbic instance.

This function is Reentrant.

Parameters

in	*xbicInstance*	: Xbic instance number

Returns

void

Precondition

6.2.3.22 Rm_GetVersionInfo()

```
void Rm_GetVersionInfo (
            Std_VersionInfoType * versioninfo )
```

This service returns the version information of this module.

This service is Non reentrant and returns the version information of this module. The version information includes:

- Module Id
- Vendor Id
- Vendor specific version numbers If source code for caller and callee of this function is available this function should be realized as a macro. The macro should be defined in the modules header file.

Parameters

out	*versioninfo*	Pointer to location to store version info

Returns

void

6.2.4 Variable Documentation

6.2.4.1 Rm_Config

```
const Rm_ConfigType Rm_Config  [extern]
```

Export RM configurations.

6.3 Mpu M7 IPV Driver

6.3.1 Detailed Description

Data Structures

- struct Mpu_M7_Ip_RegionConfigType

  Configuration structure containing the region configuration. *More...*
- struct Mpu_M7_Ip_ConfigType

  IP configuration structure. *More...*
- struct Mpu_M7_Ip_ErrorDetailsType

  Structure used to retrieve violation details. *More...*

Enum Reference

- enum Mpu_M7_Ip_CachePolicyType

  Enumeration listing the outer and inner cache policies when RASR[TEX] == 0x1XX.
- enum Mpu_M7_Ip_MemoryType

  Enumeration listing the memory and cache attributes when RASR[TEX] == 0x0XX.
- enum Mpu_M7_Ip_AccessRightsType

  Enumeration listing access permisions.
- enum Mpu_M7_Ip_ErrorType

  Enumeration listing the possible access errors.

Function Reference

- void Mpu_M7_Ip_Init (const Mpu_M7_Ip_ConfigType *pConfig)

  Initializes the Memory Protection Unit general parameters and region configurations.
- void Mpu_M7_Ip_SetRegionConfig (uint8 u8RegionNum, const Mpu_M7_Ip_RegionConfigType *const p↩ UserConfigPtr)

  Configures the region selected by u8RegionNum with the data from pUserConfigPtr.
- void Mpu_M7_Ip_Deinit (void)

  Disables the module and resets all region configurations.
- void Mpu_M7_Ip_EnableRegion (uint8 u8RegionNum, boolean bEnable)

  Enables or disabled a specific region.
- void Mpu_M7_Ip_SetAccessRight (uint8 u8RegionNum, Mpu_M7_Ip_AccessRightsType eRights)

  Modify the access rights for a specific region.
- boolean Mpu_M7_Ip_GetErrorDetails (Mpu_M7_Ip_ErrorDetailsType *pErrorDetails)

  Retrieve error details.
- void Mpu_M7_Ip_MulticoreInit (void)

  Initializes the Memory Protection Unit general parameters and region configurations.
- void Mpu_M7_Ip_Init_Privileged (const Mpu_M7_Ip_ConfigType *pConfig)

  Initializes the Memory Protection Unit general parameters and region configurations.

- void Mpu_M7_Ip_SetRegionConfig_Privileged (uint8 u8RegionNum, const Mpu_M7_Ip_RegionConfigType *const pUserConfigPtr)

  Configures the region selected by u8RegionNum with the data from pUserConfigPtr.
- void Mpu_M7_Ip_Deinit_Privileged (void)

  Disables the module and resets all region configurations.
- void Mpu_M7_Ip_EnableRegion_Privileged (uint8 u8RegionNum, boolean bEnable)

  Enables or disabled a specific region.
- void Mpu_M7_Ip_SetAccessRight_Privileged (uint8 u8RegionNum, Mpu_M7_Ip_AccessRightsType eRights)

  Modify the access rights for a specific region.
- void Mpu_M7_Ip_GetErrorRegisters (uint32 *pMmfsr, uint32 *pAddress)

  Retrieve fault information from SCB.

6.3.2 Data Structure Documentation

6.3.2.1 struct Mpu_M7_Ip_RegionConfigType

Configuration structure containing the region configuration.

Definition at line 146 of file Mpu_M7_Ip_Types.h.

Data Fields

Type	Name	Description
uint8	u8RegionNum	Region number
uint32	u32StartAddr	Memory region start address
uint32	u32EndAddr	Memory region end address
Mpu_M7_Ip_MemoryType	eMemType	Memory Type for region
Mpu_M7_Ip_AccessRightsType	eAccessRight	Access permission for region
Mpu_M7_Ip_CachePolicyType	eOuterCachePolicy	Outer Cache Policy
Mpu_M7_Ip_CachePolicyType	eInnerCachePolicy	Inner Cache Policy
uint8	u8SubRegMask	SubRegion Disable Mask
boolean	bShareable	Enables or disables shareability

6.3.2.2 struct Mpu_M7_Ip_ConfigType

IP configuration structure.

Definition at line 169 of file Mpu_M7_Ip_Types.h.

Data Fields

Type	Name	Description
Mpu_M7_Ip_CoreType	ConfigCore	Core Id number
boolean	bDefaultMapEn	Enables the default memory map background region
boolean	bEnableRunHFNMI	Enables use of the MPU when in the hard fault, non-maskable interrupt, and FAULTMASK escalated handlers
boolean	bEnMemManageInterrupt	Enable Memory Manage Fault Exception Handler
uint8	u8RegionCnt	Region Count
const Mpu_M7_Ip_RegionConfigType *	pRegionConfigArr	Region configuration array

6.3.2.3 struct Mpu_M7_Ip_ErrorDetailsType

Structure used to retrieve violation details.

Definition at line 183 of file Mpu_M7_Ip_Types.h.

Data Fields

Type	Name	Description
uint32	u32Address	Violation address
Mpu_M7_Ip_ErrorType	eType	Type of violation

6.3.3 Enum Reference

6.3.3.1 Mpu_M7_Ip_CachePolicyType

```
enum Mpu_M7_Ip_CachePolicyType
```

Enumeration listing the outer and inner cache policies when RASR[TEX] == 0x1XX.

Enumerator

MPU_M7_CACHE_POLICY_NO_CACHE	Non-cacheable
MPU_M7_CACHE_POLICY_W_BACK_WR_ALLOCATE	Write-back, write and read allocate
MPU_M7_CACHE_POLICY_W_THROUGH_NO_W_ALLOCATE	Write-through, no write allocate
MPU_M7_CACHE_POLICY_W_BACK_NO_W_ALLOCATE	Write-back, no write allocate

Definition at line 74 of file Mpu_M7_Ip_Types.h.

6.3.3.2 Mpu_M7_Ip_MemoryType

enum Mpu_M7_Ip_MemoryType

Enumeration listing the memory and cache attributes when RASR[TEX] == 0x0XX.

Enumerator

MPU_M7_MEM_STRONG_ORDER	Strongly ordered
MPU_M7_MEM_DEVICE_SHARED	Shared device
MPU_M7_MEM_NORMAL_IO_WR_THROUGH	Outer and inner write-through, no write allocate
MPU_M7_MEM_NORMAL_IO_WR_BACK1	Outer and inner write-back, no write allocate
MPU_M7_MEM_NORMAL_IO_NO_CACHE	Outer and inner Non-cacheable
MPU_M7_MEM_NORMAL_IO_WR_BACK2	Outer and inner write-back; write and read allocate
MPU_M7_MEM_DEVICE_NOSHARE	Non-shared device
MPU_M7_MEM_NORMAL_CACHEABLE	Normal Memory Cache policy enabled

Definition at line 86 of file Mpu_M7_Ip_Types.h.

6.3.3.3 Mpu_M7_Ip_AccessRightsType

enum Mpu_M7_Ip_AccessRightsType

Enumeration listing access permisions.

Enumerator

MPU_M7_PRIV_UNPRIV_NONE	0b00000000U : —|—
MPU_M7_PRIV_RW_UNPRIV_NONE	0b00000001U : rw-|—
MPU_M7_PRIV_RWX_UNPRIV_NONE	0b00010001U : rwx|—
MPU_M7_PRIV_RW_UNPRIV_R	0b00000010U : rw-|r–
MPU_M7_PRIV_RWX_UNPRIV_RX	0b00010010U : rwx|r-x
MPU_M7_PRIV_RW_UNPRIV_RW	0b00000011U : rw-|rw-
MPU_M7_PRIV_RWX_UNPRIV_RWX	0b00010011U : rwx|rwx
MPU_M7_PRIV_R_UNPRIV_NONE	0b00000101U : r–|—
MPU_M7_PRIV_RX_UNPRIV_NONE	0b00010101U : r-x|—
MPU_M7_PRIV_R_UNPRIV_R	0b00000110U : r–|r–
MPU_M7_PRIV_RX_UNPRIV_RX	0b00010110U : r-x|r-x

Definition at line 104 of file Mpu_M7_Ip_Types.h.

6.3.3.4 Mpu_M7_Ip_ErrorType

```
enum Mpu_M7_Ip_ErrorType
```

Enumeration listing the possible access errors.

Enumerator

MPU_M7_MEMMAN_ERROR_DATA_ACCESS	IACCVIOL Instruction access violation
MPU_M7_MEMMAN_ERROR_STACK	DACCVIOL Data access violation
MPU_M7_MEMMAN_ERROR_UNSTACK	MUNSTKERR MemManage fault on unstacking for a return from exception
MPU_M7_MEMMAN_ERROR_FLOATING_POINT	MSTKERR MemManage fault on stacking for exception entry
MPU_M7_MEMMAN_ERROR_MULTIPLE	MLSPERR MemManage fault during floating point lazy state preservation
MPU_M7_MEMMAN_ERROR_NONE	MemManage fault Multiple Errors detected

Definition at line 128 of file Mpu_M7_Ip_Types.h.

6.3.4 Function Reference

6.3.4.1 Mpu_M7_Ip_Init()

```
void Mpu_M7_Ip_Init (
            const Mpu_M7_Ip_ConfigType * pConfig )
```

Initializes the Memory Protection Unit general parameters and region configurations.

This function is non-reentrant

Parameters

in	*pConfig*	pointer to configuration structure for MPU M7.

Returns

void

Precondition

6.3.4.2 Mpu_M7_Ip_SetRegionConfig()

```
void Mpu_M7_Ip_SetRegionConfig (
            uint8 u8RegionNum,
            const Mpu_M7_Ip_RegionConfigType *const pUserConfigPtr )
```

Configures the region selected by u8RegionNum with the data from pUserConfigPtr.

This function is Reentrant

Parameters

in	*u8RegionNum*	Region number
in	*pUserConfigPtr*	Pointer to the region configuration structure for MPU M7.

Returns

void

Precondition

6.3.4.3 Mpu_M7_Ip_Deinit()

```
void Mpu_M7_Ip_Deinit (
            void  )
```

Disables the module and resets all region configurations.

This function is Reentrant

Returns

void

Precondition

6.3.4.4 Mpu_M7_Ip_EnableRegion()

```
void Mpu_M7_Ip_EnableRegion (
             uint8 u8RegionNum,
             boolean bEnable )
```

Enables or disabled a specific region.

This function is Reentrant

Parameters

in	*u8RegionNum*	: Region to be modified
in	*bEnable*	: Specifies wheter the region is enabled or disabled

Returns

void

Precondition

6.3.4.5 Mpu_M7_Ip_SetAccessRight()

```
void Mpu_M7_Ip_SetAccessRight (
             uint8 u8RegionNum,
             Mpu_M7_Ip_AccessRightsType eRights )
```

Modify the access rights for a specific region.

This function is Reentrant

Parameters

in	*u8RegionNum*	: Region to be modified
in	*eRights*	: Specifies the new access rights

Returns

void

Precondition

6.3.4.6 Mpu_M7_Ip_GetErrorDetails()

```
boolean Mpu_M7_Ip_GetErrorDetails (
            Mpu_M7_Ip_ErrorDetailsType * pErrorDetails )
```

Retrieve error details.

This function is Reentrant

Parameters

out	*pErrorDetails*	: Storage where the data will be saved

Returns

boolean

Return values

TRUE	if an error was present
FALSE	otherwise

Precondition

6.3.4.7 Mpu_M7_Ip_MulticoreInit()

```
void Mpu_M7_Ip_MulticoreInit (
            void  )
```

Initializes the Memory Protection Unit general parameters and region configurations.

This function is non-reentrant

Precondition

6.3.4.8 Mpu_M7_Ip_Init_Privileged()

```
void Mpu_M7_Ip_Init_Privileged (
            const Mpu_M7_Ip_ConfigType * pConfig )
```

Initializes the Memory Protection Unit general parameters and region configurations.

This function is non-reentrant

Parameters

in	*pConfig*	pointer to configuration structure for MPU M7.

Returns

void

Precondition

6.3.4.9 Mpu_M7_Ip_SetRegionConfig_Privileged()

```
void Mpu_M7_Ip_SetRegionConfig_Privileged (
            uint8 u8RegionNum,
            const Mpu_M7_Ip_RegionConfigType *const pUserConfigPtr )
```

Configures the region selected by u8RegionNum with the data from pUserConfigPtr.

This function is Reentrant

Parameters

in	*u8RegionNum*	region number
in	*pUserConfigPtr*	pointer to the region configuration structure for MPU M7.

Returns

void

Precondition

6.3.4.10 Mpu_M7_Ip_Deinit_Privileged()

```
void Mpu_M7_Ip_Deinit_Privileged (
            void  )
```

Disables the module and resets all region configurations.

This function is Reentrant

Returns

void

Precondition

6.3.4.11 Mpu_M7_Ip_EnableRegion_Privileged()

```
void Mpu_M7_Ip_EnableRegion_Privileged (
            uint8 u8RegionNum,
            boolean bEnable )
```

Enables or disabled a specific region.

This function is Reentrant

Parameters

in	*u8RegionNum*	: Region to be modified
in	*bEnable*	: Specifies wheter the region is enabled or disabled

Returns

void

Precondition

6.3.4.12 Mpu_M7_Ip_SetAccessRight_Privileged()

```
void Mpu_M7_Ip_SetAccessRight_Privileged (
            uint8 u8RegionNum,
            Mpu_M7_Ip_AccessRightsType eRights )
```

Modify the access rights for a specific region.

This function is Reentrant

Parameters

in	*u8RegionNum*	: Region to be modified
in	*eRights*	: Specifies the new access rights

Returns

void

Precondition

6.3.4.13 Mpu_M7_Ip_GetErrorRegisters()

```
void Mpu_M7_Ip_GetErrorRegisters (
             uint32 * pMmfsr,
             uint32 * pAddress )
```

Retrieve fault information from SCB.

Parameters

out	*pMmfsr*	SCB->MMFSR register contents
out	*pAddress*	Address of the violation

Returns

void

6.4 Pflash IPV Driver

6.4.1 Detailed Description

Data Structures

- struct Pflash_Ip_ConfigType

  Structure containing the configuration of the Pflash Ip. More...

Enum Reference

- enum Pflash_Ip_MasterType

  Enumeration listing Pflash master access protection field.
- enum Pflash_Ip_MasterAccessType

  Enumeration listing Pflash access type.

Function Reference

- void Pflash_Ip_Init (const Pflash_Ip_ConfigType *pPflashConfig)

  Initializes the Pflash Ip.
- void Pflash_Ip_SetAccessProtection (uint8 u8instance, Pflash_Ip_MasterType eMaster, Pflash_Ip_MasterAccessType eAccess)

  Set access protection for Pflash master.
- void Pflash_Ip_SetAccessProtection_Privileged (uint8 u8instance, Pflash_Ip_MasterType eMaster, Pflash_Ip_MasterAccessType eAccess)

  Set a master access protection by Pflash.

Variables

- const Pflash_Ip_ConfigType Pflash_Config []

  Export PFLASH configurations.

6.4.2 Data Structure Documentation

6.4.2.1 struct Pflash_Ip_ConfigType

Structure containing the configuration of the Pflash Ip.

Definition at line 116 of file Pflash_Ip_Types.h.

Data Fields

Type	Name	Description
const uint8	u32Pflash_Instance	Pflash instance
Pflash_Ip_MasterType	Pflash_Map	Pflash master access protection field
Pflash_Ip_MasterAccessType	Pflash_Access	Pflash access type

6.4.3 Enum Reference

6.4.3.1 Pflash_Ip_MasterType

```
enum Pflash_Ip_MasterType
```

Enumeration listing Pflash master access protection field.

Definition at line 81 of file Pflash_Ip_Types.h.

6.4.3.2 Pflash_Ip_MasterAccessType

```
enum Pflash_Ip_MasterAccessType
```

Enumeration listing Pflash access type.

Definition at line 104 of file Pflash_Ip_Types.h.

6.4.4 Function Reference

6.4.4.1 Pflash_Ip_Init()

```
void Pflash_Ip_Init (
            const Pflash_Ip_ConfigType * pPflashConfig )
```

Initializes the Pflash Ip.

This function is Reentrant

Parameters

in	*pPflashConfig*	Pointer for the configuration Pflash

Returns

void

Precondition

6.4.4.2 Pflash_Ip_SetAccessProtection()

```
void Pflash_Ip_SetAccessProtection (
            uint8 u8instance,
            Pflash_Ip_MasterType eMaster,
            Pflash_Ip_MasterAccessType eAccess )
```

Set access protection for Pflash master.

Parameters

in	*uint8*	u8instance - Pflash instance
in	*Pflash_Ip_MasterType*	eMaster - Pflash Master
in	*Pflash_Ip_MasterAccessType*	eAccess - Allow or not allow read Pflash

Returns

void

Precondition

6.4.4.3 Pflash_Ip_SetAccessProtection_Privileged()

```
void Pflash_Ip_SetAccessProtection_Privileged (
            uint8 u8instance,
            Pflash_Ip_MasterType eMaster,
            Pflash_Ip_MasterAccessType eAccess )
```

Set a master access protection by Pflash.

This function is Reentrant

Parameters

in	*u8instance*	Pflash instance
in	*eMaster*	Master access protection field
in	*eAccess*	Access type

Returns

void

Precondition

6.4.5 Variable Documentation

6.4.5.1 Pflash_Config

```
const Pflash_Ip_ConfigType Pflash_Config[]  [extern]
```

Export PFLASH configurations.

6.5 Sema42 IPV Driver

6.5.1 Detailed Description

Enum Reference

- enum Sema42_Ip_StatusType

  Enumeration listing the possible return codes for Sema42 Ip driver.

Function Reference

- void Sema42_Ip_Init (const uint32 u32Instance)

  Initializes the Sema42 instance by resetting all gates.
- Sema42_Ip_StatusType Sema42_Ip_LockGate (const uint32 u32Instance, const uint8 u8GateIndex, uint8 u8DomainId)

  Lock a Semaphore gate.
- Sema42_Ip_StatusType Sema42_Ip_UnlockGate (const uint32 u32Instance, const uint8 u8GateIndex)

  Unlock a Semaphore gate.
- void Sema42_Ip_ResetGate (const uint32 u32Instance, const uint8 u8GateIndex)

  Reset a Semaphore gate.
- uint8 Sema42_Ip_GetGateLocker (const uint32 u32Instance, const uint8 u8GateIndex)

  Get the current owner of a semaphore.
- boolean Sema42_Ip_IsResetGateStateIdle (const uint32 u32Instance)

  Check if the current state of the reset gate state machine is idle.
- uint8 Sema42_Ip_GetResetGateBusMaster (const uint32 u32Instance)

  Get the bus master which initiated the most recent write operation on the reset register.
- uint8 Sema42_Ip_GetResetGateIndex (const uint32 u32Instance)

  Get the index of the gate targetted by the most recent reset attempt.
- void Sema42_Ip_Init_Privileged (const uint32 u32Instance)

  Initializes the Sema42 instance by resetting all gates.
- uint8 Sema42_Ip_LockGate_Privileged (const uint32 u32Instance, const uint8 u8GateIndex, uint8 u8↩ DomainId)

  Lock a Semaphore gate.
- uint8 Sema42_Ip_UnlockGate_Privileged (const uint32 u32Instance, const uint8 u8GateIndex)

  Unlock a Semaphore gate.
- void Sema42_Ip_ResetGate_Privileged (const uint32 u32Instance, const uint8 u8GateIndex)

  Reset a Semaphore gate.
- uint8 Sema42_Ip_GetGateLocker_Privileged (const uint32 u32Instance, const uint8 u8GateIndex)

  Get the current owner of a semaphore.
- uint8 Sema42_Ip_IsResetGateStateIdle_Privileged (const uint32 u32Instance)

  Check if the current state of the reset gate state machine is idle.
- uint8 Sema42_Ip_GetResetGateBusMaster_Privileged (const uint32 u32Instance)

  Get the bus master which initiated the most recent write operation on the reset register.
- uint8 Sema42_Ip_GetResetGateIndex_Privileged (const uint32 u32Instance)

  Get the index of the gate targetted by the most recent reset attempt.

6.5.2 Enum Reference

6.5.2.1 Sema42_Ip_StatusType

```
enum Sema42_Ip_StatusType
```

Enumeration listing the possible return codes for Sema42 Ip driver.

Enumerator

Sema42_Ip_Success	Operation successful
Sema42_Ip_Error	Operation resulted in error

Definition at line 88 of file Sema42_Ip_Types.h.

6.5.3 Function Reference

6.5.3.1 Sema42_Ip_Init()

```
void Sema42_Ip_Init (
            const uint32 u32Instance )
```

Initializes the Sema42 instance by resetting all gates.

This function is non-reentrant

Parameters

in	*u32Instance*	Instance number

Returns

void

Precondition

6.5.3.2 Sema42_Ip_LockGate()

```
Sema42_Ip_StatusType Sema42_Ip_LockGate (
            const uint32 u32Instance,
```

```
            const uint8 u8GateIndex,
            uint8 u8DomainId )
```

Lock a Semaphore gate.

This function is reentrant

Parameters

in	*u32Instance*	Instance number
in	*u8GateIndex*	Index of the selected gate
in	*u8DomainId*	Current Domain of the core. If the XRDC is not supported the domain id is the core ID - Sys_GetCoreId()

Returns

Sema42_Ip_StatusType

Return values

Sema42_Ip_Success:If	the gate was locked successfully
Sema42_Ip_Error	If the gate is locked by another master

Precondition

6.5.3.3 Sema42_Ip_UnlockGate()

```
Sema42_Ip_StatusType Sema42_Ip_UnlockGate (
            const uint32 u32Instance,
            const uint8 u8GateIndex )
```

Unlock a Semaphore gate.

This function is reentrant

Parameters

in	*u32Instance*	Instance number
in	*u8GateIndex*	Index of the selected gate

Returns

Sema42_Ip_StatusType

Return values

Sema42_Ip_Success:If	the gate was unlocked successfully
Sema42_Ip_Error	If the gate is unlocked by another master

Precondition

6.5.3.4 Sema42_Ip_ResetGate()

```
void Sema42_Ip_ResetGate (
             const uint32 u32Instance,
             const uint8 u8GateIndex )
```

Reset a Semaphore gate.

This function is reentrant

Parameters

in	*u32Instance*	Instance number
in	*u8GateIndex*	Index of the selected gate

Returns

void

Precondition

6.5.3.5 Sema42_Ip_GetGateLocker()

```
uint8 Sema42_Ip_GetGateLocker (
             const uint32 u32Instance,
             const uint8 u8GateIndex )
```

Get the current owner of a semaphore.

This function is reentrant

Parameters

in	*u32Instance*	Instance number
in	*u8GateIndex*	Index of the selected gate

Returns

uint8

Return values

Index	of the semaphore owner, or SEMA42_IP_UNLOCKED_GATE if the gate is not locked.

Precondition

6.5.3.6 Sema42_Ip_IsResetGateStateIdle()

```
boolean Sema42_Ip_IsResetGateStateIdle (
            const uint32 u32Instance )
```

Check if the current state of the reset gate state machine is idle.

This function is reentrant

Parameters

in	*u32Instance*	Instance number

Returns

boolean

Return values

TRUE	if the reset gate is idle
FALSE	if the reset gate is waiting for the second write or if it is actively resseting a gate

Precondition

6.5.3.7 Sema42_Ip_GetResetGateBusMaster()

```
uint8 Sema42_Ip_GetResetGateBusMaster (
             const uint32 u32Instance )
```

Get the bus master which initiated the most recent write operation on the reset register.

This function is reentrant

Parameters

in	*u8Instance*	Instance number

Returns

uint8

Return values

Id	of the domain which requested the most recent reset

Precondition

6.5.3.8 Sema42_Ip_GetResetGateIndex()

```
uint8 Sema42_Ip_GetResetGateIndex (
             const uint32 u32Instance )
```

Get the index of the gate targetted by the most recent reset attempt.

This function is reentrant

Parameters

in	*u8Instance*	instance number

Returns

uint8 : index of the gate targeted for reset

Return values

Index	of the gate targeted for reset

Precondition

6.5.3.9 Sema42_Ip_Init_Privileged()

```
void Sema42_Ip_Init_Privileged (
            const uint32 u32Instance )
```

Initializes the Sema42 instance by resetting all gates.

This function is non-reentrant

Parameters

in	*u32Instance*	Instance number

Returns

void

Precondition

6.5.3.10 Sema42_Ip_LockGate_Privileged()

```
uint8 Sema42_Ip_LockGate_Privileged (
            const uint32 u32Instance,
            const uint8 u8GateIndex,
            uint8 u8DomainId )
```

Lock a Semaphore gate.

This function is reentrant

Parameters

in	*u32Instance*	Instance number
in	*u8GateIndex*	Index of the selected gate
in	*u8DomainId*	Current Domain of the core. If the XRDC is not supported the domain id is the core ID - Sys_GetCoreId()

Returns

uint8

Return values

Status	of Semaphore instance

Precondition

6.5.3.11 Sema42_Ip_UnlockGate_Privileged()

```
uint8 Sema42_Ip_UnlockGate_Privileged (
             const uint32 u32Instance,
             const uint8 u8GateIndex )
```

Unlock a Semaphore gate.

This function is reentrant

Parameters

in	*u32Instance*	Instance number
in	*u8GateIndex*	Index of the selected gate

Returns

uint8

Return values

Status	of Semaphore instance

Precondition

6.5.3.12 Sema42_Ip_ResetGate_Privileged()

```
void Sema42_Ip_ResetGate_Privileged (
             const uint32 u32Instance,
             const uint8 u8GateIndex )
```

Reset a Semaphore gate.

This function is reentrant

Parameters

in	*u32Instance*	Instance number
in	*u8GateIndex*	Index of the selected gate

Returns

void

Precondition

6.5.3.13 Sema42_Ip_GetGateLocker_Privileged()

```
uint8 Sema42_Ip_GetGateLocker_Privileged (
             const uint32 u32Instance,
             const uint8 u8GateIndex )
```

Get the current owner of a semaphore.

This function is reentrant

Parameters

in	*u32Instance*	Instance number
in	*u8GateIndex*	Index of the selected gate

Returns

uint8

Return values

Index	of the semaphore owner, or SEMA42_IP_UNLOCKED_GATE if the gate is not locked.

Precondition

6.5.3.14 Sema42_Ip_IsResetGateStateIdle_Privileged()

```
uint8 Sema42_Ip_IsResetGateStateIdle_Privileged (
            const uint32 u32Instance )
```

Check if the current state of the reset gate state machine is idle.

This function is reentrant

Parameters

in	*u32Instance*	Instance number

Returns

uint8

Return values

Status	of Semaphore instance

Precondition

6.5.3.15 Sema42_Ip_GetResetGateBusMaster_Privileged()

```
uint8 Sema42_Ip_GetResetGateBusMaster_Privileged (
            const uint32 u32Instance )
```

Get the bus master which initiated the most recent write operation on the reset register.

This function is reentrant

Parameters

in	*u32Instance*	instance number

Returns

uint8

Return values

Id	of the domain which requested the most recent reset

Precondition

6.5.3.16 Sema42_Ip_GetResetGateIndex_Privileged()

```
uint8 Sema42_Ip_GetResetGateIndex_Privileged (
             const uint32 u32Instance )
```

Get the index of the gate targetted by the most recent reset attempt.

This function is reentrant

Parameters

in	*u32Instance*	instance number

Returns

uint8

Return values

Index	of the gate targeted for reset

Precondition

6.6 Virt Wrapper IPV Driver

6.6.1 Detailed Description

Data Structures

- struct Virt_Wrapper_Ip_RegConfigType

  Configuration structure containing the Pin configuration for virtual wrapper. More...
- struct Virt_Wrapper_Ip_ConfigType

  Configuration structure containing the Pin configuration for virtual wrapper. More...

Enum Reference

- enum Virt_Wrapper_Ip_SlotType

  Enumeration listing pad (MSCR or IMCR) slot in a single Virtual Wrapper Parameter_n Register.
- enum Virt_Wrapper_Ip_AccessType

  Enumeration listing which PDAC will be assigned to access by the pad.

Function Reference

- void Virt_Wrapper_Ip_SetRegAccessProtection (uint32 u32Instance, Virt_Wrapper_Ip_AccessType eMirror, uint32 u32Pin, Virt_Wrapper_Ip_SlotType eSlotType)

  Virt_Wrapper_Ip_SetRegAccessProtection.

Variables

- const Virt_Wrapper_Ip_ConfigType Virt_Wrapper_Ip_Config

  Export VIRT_WRAPPER configurations.

6.6.2 Data Structure Documentation

6.6.2.1 struct Virt_Wrapper_Ip_RegConfigType

Configuration structure containing the Pin configuration for virtual wrapper.

Definition at line 121 of file Virt_Wrapper_Ip_Types.h.

6.6.2.2 struct Virt_Wrapper_Ip_ConfigType

Configuration structure containing the Pin configuration for virtual wrapper.

Definition at line 132 of file Virt_Wrapper_Ip_Types.h.

6.6.3 Enum Reference

6.6.3.1 Virt_Wrapper_Ip_SlotType

```
enum Virt_Wrapper_Ip_SlotType
```

Enumeration listing pad (MSCR or IMCR) slot in a single Virtual Wrapper Parameter_n Register.

Definition at line 99 of file Virt_Wrapper_Ip_Types.h.

6.6.3.2 Virt_Wrapper_Ip_AccessType

```
enum Virt_Wrapper_Ip_AccessType
```

Enumeration listing which PDAC will be assigned to access by the pad.

Definition at line 111 of file Virt_Wrapper_Ip_Types.h.

6.6.4 Function Reference

6.6.4.1 Virt_Wrapper_Ip_SetRegAccessProtection()

```
void Virt_Wrapper_Ip_SetRegAccessProtection (
            uint32 u32Instance,
            Virt_Wrapper_Ip_AccessType eMirror,
            uint32 u32Pin,
            Virt_Wrapper_Ip_SlotType eSlotType )
```

Virt_Wrapper_Ip_SetRegAccessProtection.

Parameters

in		

6.6.5 Variable Documentation

6.6.5.1 Virt_Wrapper_Ip_Config

```
const Virt_Wrapper_Ip_ConfigType Virt_Wrapper_Ip_Config  [extern]
```

Export VIRT_WRAPPER configurations.

6.7 XBIC IPV Driver

6.7.1 Detailed Description

Data Structures

- struct Xbic_Ip_ErrorInfoType

  Structure for error informations of Xbic Ip. More...
- struct Xbic_Ip_ConfigType

  Configuration structure for Xbic Ip. More...

Function Reference

- void Xbic_Ip_Init (const Xbic_Ip_ConfigType *pXbicConfig)

  Initialize Xbic with configuration of integrity checking for all of master/slave ports.
- void Xbic_Ip_EnableMasterFeedbackCheck (uint32 xbicInstance, uint8 masterPort, boolean bFeedback↩ CheckEnable)

  Enable/Disable feedback integrity checking for a specific master port.
- void Xbic_Ip_EnableSlaveEDCCheck (uint32 xbicInstance, uint8 slavePort, boolean bEDCCheckEnable)

  Enable/Disable attribute integrity checking for a specific slave port.
- void Xbic_Ip_FeedbackCheckAndEDCCheckDisable (void)

  Disable all of feedback integrity checking for the master ports and attribute integrity checking for slave ports.
- void Xbic_Ip_IntegrityCheckerGetErrorStatus (uint32 xbicInstance, Xbic_Ip_ErrorInfoType *pErrorInfo)

  Get error status of a Xbic instance.
- void Xbic_Ip_IntegrityErrorInjectionConfig (uint32 xbicInstance, uint8 slaveTarget, uint8 masterTargetID, uint8 synDecode)

  Inject an error into a master, a slave of a Xbic instance with a specific error code.
- void Xbic_Ip_IntegrityErrorInjectionDisable (uint32 xbicInstance)

  Disable Inject error a specific Xbic instance.
- void Xbic_Ip_WriteRegister (uint32 address, uint32 value)

  This function write into the register.
- uint32 Xbic_Ip_ReadRegister (uint32 address)

  This function return the value of register.
- void Xbic_Ip_MasterFeedbackCheck (uint32 xbicInstance, uint8 masterPort, boolean bFeedbackCheckEnable)

  Enable/Disable feedback integrity checking for a specific master port.
- void Xbic_Ip_SlaveEDCCheck (uint32 xbicInstance, uint8 slavePort, boolean bEDCCheckEnable)

  Enable/Disable attribute integrity checking for a specific slave port.
- void Xbic_Ip_LLD_IntegrityErrorInjectionConfig (uint32 xbicInstance, uint8 slaveTarget, uint8 master↩ TargetID, uint8 synDecode)

  Inject an error into a master, a slave of a Xbic instance with a specific error code.
- void Xbic_Ip_LLD_IntegrityErrorInjectionDisable (uint32 xbicInstance)

  Disable Inject error a specific Xbic instance.

Variables

- const Xbic_Ip_ConfigType Xbic_Config
  Export XBIC configurations.

6.7.2 Data Structure Documentation

6.7.2.1 struct Xbic_Ip_ErrorInfoType

Structure for error informations of Xbic Ip.

Definition at line 101 of file Xbic_Ip_Types.h.

Data Fields

Type	Name	Description
boolean	bErrorStatusValid	No Error /Error detected—all fields of the ESR and EAR registers are valid
boolean	aDataPhaseSlavePortError[(8U)]	Feedback integrity error detected on slave
boolean	aDataPhaseMasterPortError[(8U)]	Feedback integrity error detected on master
uint8	masterID	The logical master ID number of the bus master which requested the most recent transfer with an attribute integrity check error detected.
uint8	slavePort	Slave port targeted by the most recent transfer with an attribute integrity check error detected.
uint8	synError	Indicate which syndrome calculated for the most recent transfer with an attribute integrity check error detected
uint32	errAddressDetect	Indicate which address of the most recent transfer with an attribute integrity check error detected.

6.7.2.2 struct Xbic_Ip_ConfigType

Configuration structure for Xbic Ip.

Definition at line 115 of file Xbic_Ip_Types.h.

Data Fields

Type	Name	Description
uint32	u32XbicTurnCheckOnPerPort[XBIC_INSTANCE_COUNT]	This will be write on XBIC Module Control Register(MCR):to turn attribute integrity checking and feedback integrity checking on or off on a per-port basis

6.7.3 Function Reference

6.7.3.1 Xbic_Ip_Init()

```
void Xbic_Ip_Init (
            const Xbic_Ip_ConfigType * pXbicConfig )
```

Initialize Xbic with configuration of integrity checking for all of master/slave ports.

This function is Reentrant.

Parameters

in	*pXbicConfig*	: Pointer point to configuration of integrity checking for all of master/slave port.

Returns

void

Precondition

6.7.3.2 Xbic_Ip_EnableMasterFeedbackCheck()

```
void Xbic_Ip_EnableMasterFeedbackCheck (
            uint32 xbicInstance,
            uint8 masterPort,
            boolean bFeedbackCheckEnable )
```

Enable/Disable feedback integrity checking for a specific master port.

This function is Reentrant.

Parameters

in	*xbicInstance*	: Xbic instance number
in	*masterPort*	: Master port number
in	*bFeedbackCheckEnable*	: Specifies wheter the feedback integrity checking for the master port is enabled or disabled

Returns

void

Precondition

6.7.3.3 Xbic_Ip_EnableSlaveEDCCheck()

```
void Xbic_Ip_EnableSlaveEDCCheck (
            uint32 xbicInstance,
            uint8 slavePort,
            boolean bEDCCheckEnable )
```

Enable/Disable attribute integrity checking for a specific slave port.

This function is Reentrant.

Parameters

in	*xbicInstance*	: Xbic instance number
in	*slavePort*	: Slave port number
in	*bEDCCheckEnable*	: Specifies wheter the attribute integrity checking for the slave port is enabled or disabled

Returns

void

Precondition

6.7.3.4 Xbic_Ip_FeedbackCheckAndEDCCheckDisable()

```
void Xbic_Ip_FeedbackCheckAndEDCCheckDisable (
            void  )
```

Disable all of feedback integrity checking for the master ports and attribute integrity checking for slave ports.

This function is Reentrant.

Returns

void

Precondition

6.7.3.5 Xbic_Ip_IntegrityCheckerGetErrorStatus()

```
void Xbic_Ip_IntegrityCheckerGetErrorStatus (
            uint32 xbicInstance,
            Xbic_Ip_ErrorInfoType * pErrorInfo )
```

Get error status of a Xbic instance.

This function is Reentrant.

Parameters

in	*xbicInstance*	: Xbic instance number
in	*pErrorInfo*	: Pointer points to the memory area containing the XBIC reports error information

Returns

void

Precondition

6.7.3.6 Xbic_Ip_IntegrityErrorInjectionConfig()

```
void Xbic_Ip_IntegrityErrorInjectionConfig (
            uint32 xbicInstance,
            uint8 slaveTarget,
            uint8 masterTargetID,
            uint8 synDecode )
```

Inject an error into a master, a slave of a Xbic instance with a specific error code.

This function is Reentrant.

Parameters

in	*xbicInstance*	: Xbic instance number
in	*slaveTarget*	: Slave port number injected error
in	*masterTargetID*	: Master tagget ID number injected error
in	*synDecode*	: Error Syndrome injected

Returns

void

Precondition

6.7.3.7 Xbic_Ip_IntegrityErrorInjectionDisable()

```
void Xbic_Ip_IntegrityErrorInjectionDisable (
             uint32 xbicInstance )
```

Disable Inject error a specific Xbic instance.

This function is Reentrant.

Parameters

in	*xbicInstance*	: Xbic instance number

Returns

void

Precondition

6.7.3.8 Xbic_Ip_WriteRegister()

```
void Xbic_Ip_WriteRegister (
             uint32 address,
             uint32 value )
```

This function write into the register.

Parameters

in	*address*	The register address.
in	*value*	The value will be written.

6.7.3.9 Xbic_Ip_ReadRegister()

```
uint32 Xbic_Ip_ReadRegister (
            uint32 address )
```

This function return the value of register.

Parameters

in	*address*	The register address.

Returns

uint32 The value of register.

6.7.3.10 Xbic_Ip_MasterFeedbackCheck()

```
void Xbic_Ip_MasterFeedbackCheck (
            uint32 xbicInstance,
            uint8 masterPort,
            boolean bFeedbackCheckEnable )
```

Enable/Disable feedback integrity checking for a specific master port.

Parameters

in	*xbicInstance*	: Xbic instance number
in	*masterPort*	: Master port number
in	*bFeedbackCheckEnable*	: Specifies wheter the feedback integrity checking for the master port is enabled or disabled

Returns

void

6.7.3.11 Xbic_Ip_SlaveEDCCheck()

```
void Xbic_Ip_SlaveEDCCheck (
            uint32 xbicInstance,
            uint8 slavePort,
            boolean bEDCCheckEnable )
```

Enable/Disable attribute integrity checking for a specific slave port.

Parameters

in	*xbicInstance*	: Xbic instance number
in	*slavePort*	: Slave port number
in	*bEDCCheckEnable*	: Specifies wheter the attribute integrity checking for the slave port is enabled or disabled

Returns

void

6.7.3.12 Xbic_Ip_LLD_IntegrityErrorInjectionConfig()

```
void Xbic_Ip_LLD_IntegrityErrorInjectionConfig (
            uint32 xbicInstance,
            uint8 slaveTarget,
            uint8 masterTargetID,
            uint8 synDecode )
```

Inject an error into a master, a slave of a Xbic instance with a specific error code.

Parameters

in	*xbicInstance*	: Xbic instance number
in	*slaveTarget*	: Slave port number injected error
in	*masterTargetID*	: Master tagget ID number injected error
in	*synDecode*	: Error Syndrome injected

Returns

void

6.7.3.13 Xbic_Ip_LLD_IntegrityErrorInjectionDisable()

```
void Xbic_Ip_LLD_IntegrityErrorInjectionDisable (
             uint32 xbicInstance )
```

Disable Inject error a specific Xbic instance.

Parameters

in	*xbicInstance*	: Xbic instance number

Returns

void

Precondition

6.7.4 Variable Documentation

6.7.4.1 Xbic_Config

```
const Xbic_Ip_ConfigType Xbic_Config  [extern]
```

Export XBIC configurations.

6.8 XRDC IPV Driver

6.8.1 Detailed Description

Data Structures

- struct Xrdc_Ip_MemConfigType

  Configuration structure containing XRDC memory region configuration. *More...*
- struct Xrdc_Ip_PeripheralConfigType

  Configuration structure containing XRDC peripheral slot configuration. *More...*
- struct Xrdc_Ip_ErrorStatusType

  Structure used to retrieve violation details. *More...*
- struct Xrdc_Ip_DomainIDErrStatusType

  Structure used to retrieve information violation details and the domain ID where violation occured. *More...*
- struct Xrdc_Ip_DomainConfigType

  Configuration structure containing XRDC domain configuration. *More...*
- struct Xrdc_Ip_InstanceConfigType

  IP configuration structure. *More...*
- struct Xrdc_Ip_ConfigType

  IP configuration structure. *More...*

Enum Reference

- enum Xrdc_Ip_MDACInstanceType

  Enumeration listing MDAC instances.
- enum Xrdc_Ip_SecureAttributeType

  Enumeration listing secure attributes of a XRDC master.
- enum Xrdc_Ip_PriviledgedAttributeType

  Enumeration listing priviledge attributes of a XRDC master.
- enum Xrdc_Ip_DomainIDType

  Enumeration listing XRDC domain IDs.
- enum Xrdc_Ip_MasterCoreType

  Enumeration listing XRDC master core.
- enum Xrdc_Ip_MasterType

  Enumeration listing XRDC masters type.
- enum Xrdc_Ip_ErrorStateType

  Enumeration listing states of access violations.
- enum Xrdc_Ip_ErrorAccessType

  Enumeration listing errors occurred on a read or write access.
- enum Xrdc_Ip_ErrorAttributeType

  Enumeration listing error attributes of access violations.

Function Reference

- void Xrdc_Ip_Init (Xrdc_Ip_ConfigType const *pXrdcConfig)

  Initializes Extended Resource Domain Controller parameters.
- void Xrdc_Ip_InstanceInit (Xrdc_Ip_InstanceConfigType const *pXrdcInstanceConfig)

  Initializes Extended Resource Domain Controller Instance parameters.
- uint32 Xrdc_Ip_GetDomainID (uint32 u32Instance)

  Get domain ID of the current bus master making the transaction request.
- void Xrdc_Ip_SetProcessID (Xrdc_Ip_MasterCoreType MasterCore, uint8 u8Pid, Xrdc_Ip_SecureAttributeType eSecureAttr)

  Assign Process ID value to a core master.
- void Xrdc_Ip_GetDomainIDErrorStatus (uint32 u32Instance, Xrdc_Ip_DomainIDErrStatusType *pXrdc↩ DomainIDErrorStatus)

  Retrieve error details.
- void Xrdc_Ip_Init_Privileged (Xrdc_Ip_ConfigType const *pXrdcConfig)

  Initializes Extended Resource Domain Controller parameters.
- uint32 Xrdc_Ip_GetDomainID_Privileged (uint32 u32Instance)

  Get domain ID of the current bus master making the transaction request.
- void Xrdc_Ip_GetDomainIDErrorStatus_Privileged (uint32 u32Instance, Xrdc_Ip_DomainIDErrStatusType *pXrdcDomainIDErrorStatus)

  Retrieve error details.
- void Xrdc_Ip_InstanceInit_Privileged (Xrdc_Ip_InstanceConfigType const *pXrdcConfig)

  Initializes Extended Resource Domain Controller Instance parameters.
- void Xrdc_Ip_SetProcessID_Privileged (uint8 u8MasterCore, uint8 u8Pid, Xrdc_Ip_SecureAttributeType eSecureAttr)

  Assign Process ID value to a core master.

Variables

- const Xrdc_Ip_ConfigType Xrdc_Config

  Export XRDC configurations.

6.8.2 Data Structure Documentation

6.8.2.1 struct Xrdc_Ip_MemConfigType

Configuration structure containing XRDC memory region configuration.

Definition at line 305 of file Xrdc_Ip_Types.h.

Data Fields

Type	Name	Description
const uint32	u32XrdcMrcInstance	Corresponding MRC instance of current memory region.
const uint32	u32XrdcMrcRegionDescriptor	Selection of descriptor for current memory region.
const uint32	u32XrdcStartAddress	Start address of current memory region.
const uint32	u32XrdcEndAddress	End address of current memory region.
const uint32	u32XrdcSema4Enable	Enable or disable Semaphore support.
const uint32	u32XrdcSema4Number	Semaphore number used in access evaluation.

6.8.2.2 struct Xrdc_Ip_PeripheralConfigType

Configuration structure containing XRDC peripheral slot configuration.

Definition at line 320 of file Xrdc_Ip_Types.h.

Data Fields

Type	Name	Description
const uint32	u32XrdcPdacInstance	Corresponding PDAC instance of current peripheral slot.
const uint32	u32XrdcSema4Enable	Enable or disable Semaphore support.
const uint32	u32XrdcSema4Number	Semaphore number used in access evaluation.
const uint32	u32XrdcPerPolicy	Access policy of current peripheral slot.
const uint32	u32XrdcPerPolicy1	

6.8.2.3 struct Xrdc_Ip_ErrorStatusType

Structure used to retrieve violation details.

Definition at line 333 of file Xrdc_Ip_Types.h.

Data Fields

Type	Name	Description
uint32	u32AddError	Address of an access violation.
uint32	u32AddErrorRemain	
Xrdc_Ip_ErrorStateType	ErrState	State of access violations.
uint32	u32ErrPort	Port number of the MRC that detected the access violation.
Xrdc_Ip_ErrorAccessType	ErrAccess	Whether the captured access violation occurred on a read or write access.
Xrdc_Ip_ErrorAttributeType	ErrAttribute	Attributes of the access violation.

6.8.2.4 struct Xrdc_Ip_DomainIDErrStatusType

Structure used to retrieve information violation details and the domain ID where violation occured.

Definition at line 347 of file Xrdc_Ip_Types.h.

Data Fields

- Xrdc_Ip_DomainIDType DomainIDAccessError
  Domain ID of the access violation.
- Xrdc_Ip_ErrorStatusType ErrorStatus [(uint32)(19UL)]
  Error status of the access violation.

6.8.2.4.1 Field Documentation

6.8.2.4.1.1 DomainIDAccessError `Xrdc_Ip_DomainIDType DomainIDAccessError`

Domain ID of the access violation.

Definition at line 349 of file Xrdc_Ip_Types.h.

6.8.2.4.1.2 ErrorStatus `Xrdc_Ip_ErrorStatusType ErrorStatus[(uint32)(19UL)]`

Error status of the access violation.

Definition at line 350 of file Xrdc_Ip_Types.h.

6.8.2.5 struct Xrdc_Ip_DomainConfigType

Configuration structure containing XRDC domain configuration.

Definition at line 356 of file Xrdc_Ip_Types.h.

Data Fields

Type	Name	Description
Xrdc_Ip_DomainIDType	u32XrdcDomainID	XRDC domain ID.
Xrdc_Ip_MDACInstanceType	XrdcMdacInstance	Corresponding MDA instance of current master.
Xrdc_Ip_MasterType	XrdcCoreMdacInstance	Core or noncore attribute of current master.
const uint32	u32XrdcProcessID	PID value of curent domain ID.
const uint32	u32XrdcProcessIDMask	PIDM value of current domain ID.
const uint32	u32XrdcProcessIDEnable	Enable or disable PID support.
const uint32	u32XrdcThreeStateModel	If core master support three-state model or not.
const uint32	u32XrdcWordDescriptor	Selection of descriptor for current core master.
const uint32	u32XrdcDIDBypassDisable	Enable or disable DID bypass.
Xrdc_Ip_SecureAttributeType	eXrdcSecureOutput	Secure attribute of current master.
Xrdc_Ip_PriviledgedAttributeType	eXrdcPriviledgeOutput	Priviledged attribute of current master.

6.8.2.6 struct Xrdc_Ip_InstanceConfigType

IP configuration structure.

Definition at line 377 of file Xrdc_Ip_Types.h.

Data Fields

Type	Name	Description
const uint32	u32XrdcInstance	XRDC instance.
const Xrdc_Ip_DomainConfigType *	pDomainConfig	Pointer to a list of configured domains.
const uint32	u32DomainConfigCnt	Number of configured domains.
const Xrdc_Ip_MemConfigType *	pMemoryDesConfig	Pointer to a list of configured memory regions.
const uint32	u32MemoryConfigCnt	Number of configured memory regions.
const Xrdc_Ip_PeripheralConfigType *	pPeripheralDescriptorConfig	Pointer to a list of configured peripheral slots.
const uint32	u32PeripheralConfigCnt	Number of configured peripheral slots.

6.8.2.7 struct Xrdc_Ip_ConfigType

IP configuration structure.

Definition at line 391 of file Xrdc_Ip_Types.h.

Data Fields

Type	Name	Description
const Xrdc_Ip_InstanceConfigType *const *const	pInstanceConfig	Pointer to a list of configured instances.
const uint32	u32InstanceConfigCnt	Number of configured instances.
const uint32 *const	pInstanceInUsed	Pointer to a list of configured XRDC instances.
const uint32	u32InstanceInUsedCnt	Number of configured XRDC instances.

6.8.3 Enum Reference

6.8.3.1 Xrdc_Ip_MDACInstanceType

```
enum Xrdc_Ip_MDACInstanceType
```

Enumeration listing MDAC instances.

Definition at line 152 of file Xrdc_Ip_Types.h.

6.8.3.2 Xrdc_Ip_SecureAttributeType

enum Xrdc_Ip_SecureAttributeType

Enumeration listing secure attributes of a XRDC master.

Definition at line 198 of file Xrdc_Ip_Types.h.

6.8.3.3 Xrdc_Ip_PriviledgedAttributeType

enum Xrdc_Ip_PriviledgedAttributeType

Enumeration listing priviledge attributes of a XRDC master.

Definition at line 208 of file Xrdc_Ip_Types.h.

6.8.3.4 Xrdc_Ip_DomainIDType

enum Xrdc_Ip_DomainIDType

Enumeration listing XRDC domain IDs.

Definition at line 218 of file Xrdc_Ip_Types.h.

6.8.3.5 Xrdc_Ip_MasterCoreType

enum Xrdc_Ip_MasterCoreType

Enumeration listing XRDC master core.

most significant byte is XRDC instance. least significant byte is Master core number. Ex: 0x110 - XRDC instance 1, Master core number 16.

Definition at line 245 of file Xrdc_Ip_Types.h.

6.8.3.6 Xrdc_Ip_MasterType

enum Xrdc_Ip_MasterType

Enumeration listing XRDC masters type.

Definition at line 254 of file Xrdc_Ip_Types.h.

6.8.3.7 Xrdc_Ip_ErrorStateType

enum Xrdc_Ip_ErrorStateType

Enumeration listing states of access violations.

Definition at line 265 of file Xrdc_Ip_Types.h.

6.8.3.8 Xrdc_Ip_ErrorAccessType

enum Xrdc_Ip_ErrorAccessType

Enumeration listing errors occurred on a read or write access.

Definition at line 276 of file Xrdc_Ip_Types.h.

6.8.3.9 Xrdc_Ip_ErrorAttributeType

enum Xrdc_Ip_ErrorAttributeType

Enumeration listing error attributes of access violations.

Definition at line 287 of file Xrdc_Ip_Types.h.

6.8.4 Function Reference

6.8.4.1 Xrdc_Ip_Init()

```
void Xrdc_Ip_Init (
            Xrdc_Ip_ConfigType const * pXrdcConfig )
```

Initializes Extended Resource Domain Controller parameters.

Parameters

in	*pXrdcConfig*	Pointer for configuration Xrdc

Returns

void

6.8.4.2 Xrdc_Ip_InstanceInit()

```
void Xrdc_Ip_InstanceInit (
            Xrdc_Ip_InstanceConfigType const * pXrdcInstanceConfig )
```

Initializes Extended Resource Domain Controller Instance parameters.

Parameters

in	*pXrdcConfig*	Pointer for configuration Xrdc

Returns

void

6.8.4.3 Xrdc_Ip_GetDomainID()

```
uint32 Xrdc_Ip_GetDomainID (
            uint32 u32Instance )
```

Get domain ID of the current bus master making the transaction request.

Parameters

in	*u32Instance*	XRDC Instance number

Returns

uint32

Return values

DID	of the bus master making the current transaction request

Precondition

6.8.4.4 Xrdc_Ip_SetProcessID()

```
void Xrdc_Ip_SetProcessID (
            Xrdc_Ip_MasterCoreType MasterCore,
            uint8 u8Pid,
            Xrdc_Ip_SecureAttributeType eSecureAttr )
```

Assign Process ID value to a core master.

Parameters

in	*MasterCore*	Index of configured core master
in	*u8Pid*	Process ID value
in	*eSecureAttr*	Secure attribute of configured core master

Returns

void

6.8.4.5 Xrdc_Ip_GetDomainIDErrorStatus()

```
void Xrdc_Ip_GetDomainIDErrorStatus (
            uint32 u32Instance,
            Xrdc_Ip_DomainIDErrStatusType * pXrdcDomainIDErrorStatus )
```

Retrieve error details.

This function implement errata workaround for safety measure of all platform

Parameters

in	*u8Instance*	XRDC Instance number
out	*pXrdcDomainIDErrorStatus*	Pointer getting error status
out		

6.8.4.6 Xrdc_Ip_Init_Privileged()

```
void Xrdc_Ip_Init_Privileged (
            Xrdc_Ip_ConfigType const * pXrdcConfig )
```

Initializes Extended Resource Domain Controller parameters.

Parameters

in	*pXrdcConfig*	Pointer for configuration Xrdc

Returns

void

6.8.4.7 Xrdc_Ip_GetDomainID_Privileged()

```
uint32 Xrdc_Ip_GetDomainID_Privileged (
            uint32 u32Instance )
```

Get domain ID of the current bus master making the transaction request.

Parameters

in	*u32Instance*	XRDC Instance number

Returns

uint32

Return values

DID	of the bus master making the current transaction request

Precondition

6.8.4.8 Xrdc_Ip_GetDomainIDErrorStatus_Privileged()

```
void Xrdc_Ip_GetDomainIDErrorStatus_Privileged (
            uint32 u32Instance,
            Xrdc_Ip_DomainIDErrStatusType * pXrdcDomainIDErrorStatus )
```

Retrieve error details.

This function implement errata workaround for safety measure of all platform

Parameters

in	*u8Instance*	XRDC Instance number
out	*pXrdcDomainIDErrorStatus*	Pointer getting error status
out		

6.8.4.9 Xrdc_Ip_InstanceInit_Privileged()

```
void Xrdc_Ip_InstanceInit_Privileged (
            Xrdc_Ip_InstanceConfigType const * pXrdcConfig )
```

Initializes Extended Resource Domain Controller Instance parameters.

Parameters

in	*pXrdcConfig*	Pointer for configuration Xrdc

Returns

void

6.8.4.10 Xrdc_Ip_SetProcessID_Privileged()

```
void Xrdc_Ip_SetProcessID_Privileged (
            uint8 u8MasterCore,
            uint8 u8Pid,
            Xrdc_Ip_SecureAttributeType eSecureAttr )
```

Assign Process ID value to a core master.

Parameters

in	*u8MasterCore*	Index of configured core master
in	*u8Pid*	Process ID value
in	*eSecureAttr*	Secure attribute of configured core master

Returns

void

6.8.5 Variable Documentation

6.8.5.1 Xrdc_Config

```
const Xrdc_Ip_ConfigType Xrdc_Config  [extern]
```

Export XRDC configurations.

How to Reach Us:

Home Page:
nxp.com

Web Support:
nxp.com/support

Information in this document is provided solely to enable system and software implementers to use NXP products. There are no express or implied copyright licenses granted hereunder to design or fabricate any integrated circuits based on the information in this document. NXP reserves the right to make changes without further notice to any products herein.

NXP makes no warranty, representation, or guarantee regarding the suitability of its products for any particular purpose, nor does NXP assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation consequential or incidental damages. "Typical" parameters that may be provided in NXP data sheets and/or specifications can and do vary in different applications, and actual performance may vary over time. All operating parameters, including "typicals," must be validated for each customer application by customer's technical experts. NXP does not convey any license under its patent rights nor the rights of others. NXP sells products pursuant to standard terms and conditions of sale, which can be found at the following address: nxp.com/SalesTermsandConditions.

NXP, the NXP logo, NXP SECURE CONNECTIONS FOR A SMARTER WORLD, COOLFLUX, EMBRACE, GREENCHIP, HITAG, I2C BUS, ICODE, JCOP, LIFE VIBES, MIFARE, MIFARE CLASSIC, MIFARE DESFire, MIFARE PLUS, MIFARE FLEX, MANTIS, MIFARE ULTRALIGHT, MIFARE4MOBILE, MIGLO, NTAG, ROADLINK, SMARTLX, SMARTMX, STARPLUG, TOPFET, TRENCHMOS, UCODE, Freescale, the Freescale logo, AltiVec, C–5, CodeTEST, CodeWarrior, ColdFire, ColdFire+, C–Ware, the Energy Efficient Solutions logo, Kinetis, Layerscape, MagniV, mobileGT, PEG, PowerQUICC, Processor Expert, QorIQ, QorIQ Qonverge, Ready Play, SafeAssure, the SafeAssure logo, StarCore, Symphony, VortiQa, Vybrid, Airfast, BeeKit, BeeStack, CoreNet, Flexis, MXC, Platform in a Package, QUICC Engine, SMARTMOS, Tower, TurboLink, and UMEMS are trademarks of NXP B.V. All other product or service names are the property of their respective owners. ARM, AMBA, ARM Powered, Artisan, Cortex, Jazelle, Keil, SecurCore, Thumb, TrustZone, and Vision are registered trademarks of ARM Limited (or its subsidiaries) in the EU and/or elsewhere. ARM7, ARM9, ARM11, big.LITTLE, CoreLink, CoreSight, DesignStart, Mali, mbed, NEON, POP, Sensinode, Socrates, ULINK and Versatile are trademarks of ARM Limited (or its subsidiaries) in the EU and/or elsewhere. All rights reserved. Oracle and Java are registered trademarks of Oracle and/or its affiliates. The Power Architecture and Power.org word marks and the Power and Power.org logos and related marks are trademarks and service marks licensed by Power.org.

© 2022 NXP B.V.